MATH INQUIRERS HOLIDAY PROGRAM

# Math Inquirers P6 Tutor Guide

Holiday program for the Primary 6 level

**Notice it. Test it. Explain why.**

**PRIMARY 6 HOLIDAY PROGRAM**

## Math Inquirers Holiday Program

*Notice it. Test it. Explain why.*

Program guide for tutors: schedule, lesson routine, ten minute-by-minute lesson scripts with answer keys, a progress tracker and a parent summary. Math Inquirers is online first: every lesson runs live on video; the face-to-face notes are kept for partner centres.

### The program at a glance

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|---|---|
| **Level** | Primary 6 students after the PSLE (ages 11 to 13), going into Secondary 1, who enjoy thinking and coding puzzles |
| **Length** | 10 lessons of 90 minutes, two a week over the 5 weeks of the holidays |
| **Group size** | Up to 8 children per tutor, live online. Pilot with 6. Add a second tutor or helper above 8. |
| **Focus** | Thinking skills: making a list in order, guess and check, drawing a model, working backwards, finding patterns, reasoning and explaining. Not speed drills. |
| **The arc of every lesson** | Start concrete with objects, record in a table, spot the pattern, state the rule, use it (often backwards), then explain why. |
| **What children take home** | A student workbook with an inquiry journal, 10 worksheets and a certificate |
| **Before tutors start** | Every tutor signs the tutor confidentiality agreement before receiving any materials. |

### Schedule

| Week | Lesson | Title | Topic | Big idea |
|---|---|---|---|---|
| Week 1a | 1 | **Code the Pattern** | Sequences and the nth Term | Pattern spotting |
| Week 1b | 2 | **Prime Hunters** | Primes, Factors and Prime Factorisation | Systematic listing (factor pairs) |

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| Week 2a | 3 | **Below Zero** | Negative Numbers | Number line reasoning |
| Week 2b | 4 | **Algebra Machines** | Expressions and Equations | Rule finding from a table |
| Week 3a | 5 | **Every Possible Way** | Counting | Systematic listing |
| Week 3b | 6 | **Fair or Unfair?** | Probability | Listing equally likely outcomes |
| Week 4a | 7 | **Turtle Geometry** | Angles in Polygons | Turn (exterior) and inside (interior) angles |
| Week 4b | 8 | **Secret Codes** | Clock (Modular) Arithmetic | Remainders and mod |
| Week 5a | 9 | **Data Detectives** | Averages and Data | Mean |
| Week 5b | 10 | **Showcase Challenge** | Mixed: Maths and Code | Choosing a strategy |

Suggested pairing: two lessons a week, for example Tuesday and Thursday, or Saturday and Sunday. Keep the same time slot all five weeks.

## The 90-minute lesson routine

Every lesson uses the same routine so children know what to expect and tutors can prepare quickly.

| Minutes | Part | What happens | Face to face | Online |
|---|---|---|---|---|
| **0–2** | **Opening** | Welcome, the hook and today's question. | Sit in a circle or horseshoe. | Cameras on. |
| **2–5** | **Sneak-peek puzzles** | Two puzzles on ideas we have not learnt yet. The first is an access puzzle any child can try with objects or a drawing. Answers are shown to the camera at the same moment, not read out. Nobody is marked. | Own paper held to the tutor. | Everyone holds their paper to the camera at the same moment, or shows it to the tutor only on camera. |
| **5–10** | **Warm-up** | A quick game (5 to 8), then quick questions (8 to 10). Everyone joins in. | Sit in a circle or horseshoe. | Cameras on. Answers by thumbs, hands or chat. |
| **10–25** | **Explore** | Hands-on in pairs. Objects first, no writing yet. | Materials box on each table. | Each child has their own materials at home. Pairs go to breakout rooms. |
| **25–35** | **Mini-lesson** | Tutor builds the table on the board, names the rule and asks what stays the same and what changes. | Whiteboard. | Shared slide and annotation tool. |
| **35–40** | **Brain break** | Stand, stretch, drink water. | A short movement game. | Stand up on camera. Count the stretches together. |
| **40–75** | **Worksheet** | Start in pairs, finish alone. Tutor asks "how do you know?" rather than giving answers. | Printed workbook, tutor circulates. | Printed workbook. Child holds the page up to the camera. Tutor checks one or two answers each. |
| **75–82** | **Talk time** | Volunteers share an answer and explain it. Pass is always fine. No cold-calling by name, no tallies or rankings. | Volunteers first. | Volunteers unmute. Invitations to try Think harder go to the whole group. |

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| **82–90** | **Exit slip, looking back, stamp** | 82 to 86 exit slip, 86 to 88 looking back (2 minutes, private), 88 to 90 close and stamp. | Paper slip, tutor stamps. | Child writes the answer on the Finish page and shows it to the tutor only. Tutor stamps the next time or posts it. |

## Teaching habits that matter

- Ask “How do you know?” after every answer, right or wrong.
- Wait 10 seconds before helping. Let children struggle a little.
- Praise the thinking (“You checked it two ways”), not just the answer.
- Let children disagree with each other and settle it with objects or a drawing.
- If a child finishes early, point to Think harder, Go further or the Investigation on the Finish page, not extra drill. Never say who is on which page, and never split a pair unless both children agree.

## Preparing the materials

- Send the workbook PDF to parents one week before. Ask them to print it single-sided and the cut-out pages on card if they can.
- Each child needs a computer or tablet with Scratch 3 (free online). Python is optional (any online Python editor). Cut-outs used: unplugged code cards (Lessons 1, 4, 7), square grid (Lessons 7, 9).
- Keep the answer key with the tutor only. Check answers together on screen.
- Use the shared screen or a digital whiteboard for the mini-lesson table.

## Setting up online

- Send parents the workbook and materials list one week before. Needed: the printed workbook and cut-outs, scissors, a ruler, paper and a pencil, and a computer with Scratch.
- Use one approved platform with breakout rooms and screen share. Admit only registered children and never share the link publicly.
- Ask the parent to sit nearby but not give answers. The camera should show the child’s page and hands.
- Hold up the answer page to the camera at the end of each lesson, or take a photo and send it through the approved channel.
- Do not record lessons (this is in the tutor agreement). Add a second tutor or helper above 8 children.

## Materials master list

| Lesson | Title | Face to face | Online set-up |
|---|---|---|---|
| **1** | Code the Pattern | Laptop or desktop with Scratch 3 open (scratch.mit.edu, no account needed), about 25 matchsticks or toothpicks per child (or squared paper), mini whiteboard, workbook. Optional: a Python editor for children who already know Python. | The tutor shares the slides and a Scratch window. Share your Scratch window and build the four blocks together on screen; the blocks are also on the slide. Block colours on our slides are simplified; in Scratch, find blocks by the category name on the left, not by colour. Scratch is free at scratch.mit.edu and needs no account in class. Children hold paper up to the camera or share their screen when asked. Matchsticks can be drawn as lines. |
| **2** | Prime Hunters | Laptop or desktop with Scratch 3 open, a printed 1 to 100 grid (or squared paper), two coloured pencils, mini whiteboard, workbook. Optional: a Python editor for children who already know Python. | Share the 1 to 100 grid on screen and annotate it while children work on their own printed or hand-drawn grid. Build the prime checker together in your shared Scratch window; the blocks are also on the slide. Block colours on our slides are simplified; in Scratch, find blocks by the category name on the left, not by colour. Children hold their grid up to the camera at minute 22. |

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| **3** | Below Zero | Laptop or desktop with Scratch 3 (scratch.mit.edu, free, no log-in needed), mini whiteboard and marker, workbook, pencil. | Children keep Scratch open in one browser tab and the video call in another. The tutor screen-shares Scratch to show the x number under the stage. Children share their screens only when asked; page 1 of the worksheet is done on paper with laptops turned away. Type the minus sign with the keyboard minus key (the hyphen); do not copy and paste a number from the slide or chat, because the slide uses a different minus character that Scratch will not accept. Block colours on our slides are simplified; in Scratch, find blocks by the category name on the left, not by colour. |
| **4** | Algebra Machines | Laptop or desktop with Scratch 3 (scratch.mit.edu, free), mini whiteboard and marker, workbook, pencil. | Pairs work in breakout rooms for the explore: one partner shares their screen with the machine running and types in the numbers the other partner calls out. Page 1 of the worksheet is done on paper with laptops turned away. Block colours on our slides are simplified; in Scratch, find blocks by the category name on the left, not by colour. |
| **5** | Every Possible Way | Laptop or desktop with Scratch 3 (scratch.mit.edu) open, the starter project “Every Order” (one loop that says each tea), mini whiteboards, digit cards 0 to 9, squared paper. | Children share their screen when they test code. The tutor shares the starter project as an .sb3 file in the chat (children use File, Load from your computer; no account needed), and the three starter blocks are also on the slide. Block colours on our slides are simplified; in Scratch, find blocks by the category name on the left, not by colour. Digit cards can be cut from paper. Breakout rooms in pairs for the Explore. |

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| **6** | Fair or Unfair? | Laptop or desktop with Scratch 3 (scratch.mit.edu), two dice per pair (or the tutor's dice program), a coin, coloured counters and a paper bag, mini whiteboards, tally sheet. | Children roll real dice at home; anyone without dice builds the dice program live in the shared Scratch window (pick random 1 to 6, twice) and uses that. This lesson is enrichment: probability is new to most children, so keep it playful. Block colours on our slides are simplified; in Scratch, find blocks by the category name on the left, not by colour. Pairs play the Explore game in breakout rooms and type their tallies into the chat so the group total can be added up. |
| **7** | Turtle Geometry | A laptop or desktop with Scratch 3 open (scratch.mit.edu) and the Pen extension added, mini whiteboards, a protractor, plain or squared paper. | The tutor builds a "Turtle start" Scratch file beforehand (Pen extension added and the 4 set-up blocks already in place; no .sb3 file exists in this repository) and shares it through the approved channel, so children only open it. Pairs work in breakout rooms: one child drives Scratch and shares the screen, the other predicts, then they swap. Children hold drawings up to the camera. A tablet works for Scratch but is slower. Every drawing starts with the same 4 set-up blocks (erase all, go to x: -50 y: 100, point in direction 90, pen down) so shapes stay on the stage; turn blocks in Scratch show an arrow (turn ↻ is clockwise, our "turn right"). Block colours on our slides are simplified; in Scratch, find blocks by the category name on the left (Motion, Pen, Control), not by colour. |

| | | | |
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| **8** | Secret Codes | A laptop or desktop with Scratch 3 open (the tutor builds and shares a starter project with a list called alphabet holding A to Z; no .sb3 file exists in this repository, so build it beforehand), mini whiteboards, paper for code strips, scissors. | Children make their code strips on paper and hold them to the camera. Children hold up their coded name on paper to the camera, or the tutor reads out the coded names; partners decode on paper. Pairs use breakout rooms for the coding part; the driver shares the screen. Starter project: the tutor shares an .sb3 file (not a link only) containing the list alphabet with A to Z already filled in, and the loop already built; children add the three cipher blocks. Block colours on our slides are simplified; in Scratch, list blocks are red-orange (Variables palette) and you find blocks by the category name on the left, not by colour. |
| **9** | Data Detectives | Laptop or desktop with Scratch 3 (scratch.mit.edu; free, in the browser), workbook, pencil, mini whiteboard, about 25 linking cubes or counters for levelling towers. A calculator only for checking. | Data is collected in the chat. Children share their screen to show their code. The tutor shares a starter project as an .sb3 file (one sprite with an empty list called data; children use File, Load from your computer) or children build it from the slide in 5 minutes. Block colours on our slides are simplified; in Scratch, find blocks by the category name on the left, not by colour. The mean is familiar from primary school; the new parts are lists and loops, median and mode, and misleading averages. |

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| **10** | Showcase Challenge | Laptop or desktop with Scratch 3 (scratch.mit.edu), workbook (challenge round on page 1, planning sheet on page 2), pencil, mini whiteboard, the starter projects as .sb3 files (A Mystery Machine, B Prime or Not?, C Lucky Seven, D Clock Jumper; children use File, Load from your computer), or build them from the slides, certificates, stamps. | Challenge round in teams of 2 in breakout rooms. Building: everyone in the main room with cameras on, asking for help in the chat. Showcase: in two rounds of 4, each child shares their screen for 2 minutes 30 seconds; the audience writes one question for each presenter and the tutor posts stars first (two stars and a question). A child may show a screenshot and the tutor runs the file. Children may paste a screenshot into the chat instead and the tutor runs their file, which removes screen-share switching. Block colours on our slides are simplified; in Scratch, find blocks by the category name on the left, not by colour. |

## How to use the scripts

Rules for every lesson: no public labels (never say advanced, top group, strong or early finisher), no announcing who is on which page, no cold-calling by name, no public tallies or rankings, and “pass” is always fine. Every lesson below is a minute-by-minute script. The slides match it: one slide or two for each part. You do not have to say the words exactly, but keep the questions, because the questions are what makes children think.

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| **SAY** | Words to say aloud, in plain language for 11- to 13-year-olds. Said in your own voice, not read from the page. |
| **DO** | What you do: set up, hand out, show a slide, or move around the room. |
| **ASK / EXPECT** | Questions to put to the group, with the answer you are listening for. Let children answer before you explain. |
| **IF STUCK** | A ladder of help. Give hint 1, wait ten seconds, then give hint 2. Do not give the answer. |

### Five habits that run through every script

- Predict first. Before children test anything, ask for a guess and write it down. A wrong guess is useful because we can test it.
- Wait ten seconds after every question. Count silently. Most children answer in the last four.
- Ask “How do you know?” after right answers too, so that children do not read your face for the answer.
- Praise the thinking, not the answer: “You checked it two ways” beats “Well done”.
- When an answer is wrong, say “Let us test that,” and use objects, not “No”.

### Answering a wrong answer

| If... | Then... |
|---|---|
| A child gives a wrong answer | “Thank you. Let us test it with the counters.” Then build it together. |
| A child gives a right answer but cannot explain | “Show me with your counters.” Then: “Tell your partner how you know.” |
| A child says “I do not know” | “What do you notice? What could you try first?” Offer hint 1. |
| Two children disagree | “Both of you, convince the group using objects.” Let the objects decide. |

# Lesson 1: Code the Pattern

Week 1 (first lesson) • Sequences and the nth Term • Workbook: Lesson 1

| | |
|---|---|
| **Objective** | Continue a sequence, find the rule for the nth term of a sequence with a constant difference, use it to find far terms such as the 100th, work backwards to decide whether a number is in a sequence, and test predictions with a loop and a variable in Scratch. |
| **Thinking skills** | Pattern spotting, generalising with a letter, predicting, testing in code, debugging (counting the jumps) |
| **Materials** | Laptop or desktop with Scratch 3 open (scratch.mit.edu, no account needed), about 25 matchsticks or toothpicks per child (or squared paper), mini whiteboard, workbook. Optional: a Python editor for children who already know Python. |
| **Online set-up** | The tutor shares the slides and a Scratch window. Share your Scratch window and build the four blocks together on screen; the blocks are also on the slide. Block colours on our slides are simplified; in Scratch, find blocks by the category name on the left, not by colour. Scratch is free at scratch.mit.edu and needs no account in class. Children hold paper up to the camera or share their screen when asked. Matchsticks can be drawn as lines. |
| **Prepare** | Solve the worksheet yourself first. Set out one materials box per pair. Open the slides for this lesson and check the answers in the speaker notes. |

### 0–2 min Opening

**SAY** *"Welcome to Math Inquirers P6! The PSLE is done, so there is no exam today: just puzzles, patterns and code."*

*"Every lesson we notice a pattern, test it in code, and explain why. Today's question: how can you find the 100th number in a pattern without writing out all 100?"*

**DO** Greet each child by name. Show the lesson title slide. Read "Today we will…" aloud.

### 2–5 min Sneak-peek puzzles

**SAY** *"Two sneak-peek puzzles. They are ideas we have not learnt yet, so guessing is welcome and "I do not know" is welcome. Nobody is marked. Write your answers on your own paper. When I say "show", everyone holds their paper to the camera at the same moment."*

| Sneak-peek puzzle | Answer and what to listen for |
|---|---|
| Matchsticks make triangles in a row: 1 triangle uses 3 sticks, 2 triangles use 5 sticks, 3 triangles use 7 sticks. Draw or build 4 triangles in a row. How many sticks? | 9 sticks. Each new triangle adds 2 more sticks (3, 5, 7, 9). Any child can draw it and count. |
| A taxi charges $3 to start plus $2 for every km. What is the fare for 20 km? | $43. Each km adds $2, so 2 × 20 + 3 = 43. (Not 2 × 20 = 40: the $3 start charge is there from the beginning.) A guess or "I do not know" is fine; we will learn how today. |

**DO** Puzzle 1 is an access puzzle: any child can try it with objects, a drawing or counting, so everyone gets a first win. Puzzle 2 is a thinking puzzle on the idea of today's lesson.

Papers are shown to the camera by everyone at the same moment, or to you only on camera. Never use private chat and never read wrong answers aloud. Reveal the answers to everyone at minute 5 and say: "Both of these are what we learn today."

Do not record the sneak-peek as a mark. Never announce who got which right, never say who will start on which page, and never use the word "advanced" or "top group".

Invite the WHOLE group, never one child: "When you have finished Start here, you can choose Think harder, Go further or the Investigation." Nobody is singled out. The point printed on page 1 ("After question N you may choose…") is the same for all. Any child can choose it, and any child can go back. Do not split a pair unless both children agree.

### 5–10 min Warm-up

**DO** Next! Pattern quick-fire. Six sequence cards. Pairs type the next two terms in the chat and say the rule if they spot it. No racing and no points. (5 to 8 minutes. Keep it friendly: no racing or timers, and no points for being first.)

**SAY** *"Pairs. I will show a sequence card. Take your time and type the next two terms in the chat. If you spot the rule, say it. There is no race and there are no points."*

*"Careful: not every card adds the same amount each time."*

**DO** At about minute 8, ask the quick questions on the warm-up slides. Use thumbs or a whisper to partners before taking answers. Stop at minute 10.

| Ask | Expect |
|---|---|
| 4, 9, 14, 19, __ | 24 |
| The 10th term of 2, 4, 6, 8, … | 20 |
| The 10th term of 3, 5, 7, 9, … | 21 |
| Ali has $12 and saves $5 a week. How much does he have after 6 weeks? | $42 |
| 100, 93, 86, __ | 79 |

### 10–25 min Explore (in pairs, with objects)

**SAY** *"Pattern 1 is a triangle made of 3 matchsticks. Each new pattern adds one more triangle in the row. Before you build anything: how many sticks will pattern 100 need? Write your guess in the corner of your whiteboard."*

**DO** 1. Build matchstick triangles in a row: pattern 1 has 1 triangle, pattern 2 has 2, pattern 3 has 3.

2. Count the sticks in each pattern. Make a table.

3. Predict the sticks in pattern 10 and in pattern 100. Do not build them!

4. Check pattern 10 in Scratch with a loop and a variable.

**SAY** *"Talk to your partner while you work. I will come round and ask you questions."*

**IF STUCK** Stuck building: lay the triangles point up, point down, point up, so each new triangle shares a stick.

Stuck on pattern 100: "How many sticks does each new triangle add? How many new triangles come after the first one?"

Finished early: in Scratch, set sticks to 3, repeat 9, change sticks by 2, say sticks. Does it match your pattern 10? Now change it to reach pattern 100.

**DO** Go further (from minute 12, on the explore slide, for any pair that wants it): What if each shape in the row were a SQUARE, sharing one side with the next? Predict how many sticks 10 squares in a row need. Then test pattern 3 by drawing. Answer: 31 sticks. The first square uses 4 sticks and each new square adds 3: $4 + 3 \times 9 = 31$, or $3 \times 10 + 1$. Nobody has to wait: a pair that has finished can pick this up.

**DO** At about minute 22, say "Pencils down" and bring the group back together. Take the three questions below, one at a time.

| Ask | Expect |
|---|---|
| How many sticks are in pattern 10? | 21 |
| How many in pattern 100? How did you get it without building it? | 201. Each new triangle adds 2 sticks: $3 + 2 \times 99 = 201$, or $2 \times 100 + 1$. |
| Which pattern uses exactly 75 sticks? | Pattern 37: $2 \times 37 + 1 = 75$. |

**25–35 min Mini-lesson (you teach, they join in)**

**DO** Show the mini-lesson slides. Think aloud as you work. Say what you notice, what you test and how you check. Children copy how you think.

**SAY** *"5, 8, 11, 14. The term-to-term rule is add 3. That is great for the next term, but slow for the 100th: I would have to add 3 ninety-nine times."*

*"Adding 3 each time is like the 3 times table: 3, 6, 9, 12. Each term is 2 more than the times-table number. So the nth term is $3 \times n + 2$, which we write $3n + 2$. Check the 4th term: $3 \times 4 + 2 = 14$. Yes."*

*"So the 100th term is $3 \times 100 + 2 = 302$. Now I test it in code. I set term to 5, the 1st term. From the 1st term to the 100th term there are 99 jumps, so repeat 99, change term by 3, then say term. It says 302. The maths and the code agree."*

| Worked example on the slides | Answer |
|---|---|
| | |

| | |
|---|---|
| The 100th term of 5, 8, 11, 14, .... The difference is 3, so compare with the 3 times table. Each term is 2 more: nth term = 3n + 2. Check the 4th term: 3 × 4 + 2 = 14. | 3 × 100 + 2 = 302. |
| Test it in code. set term to 5 (the 1st term). repeat 99: change term by 3 (99 jumps from the 1st term to the 100th). say term. | It says 302, the same as the rule. |

**SAY** *"The key idea today is: the rule for the nth term. The difference tells you the times table. Compare with it to find what to add or take away. Say it with me."*

**DO** Show the key idea slide at minutes 33 to 35. The "common mistakes" on that slide are for you, not the children. Watch for them during the worksheet:

- Using only the difference as the rule (writing 3n for 5, 8, 11, ...) and not comparing with the 1st term.
- Off by one: doing 100 jumps from the 1st term (5 + 3 × 100) instead of 99, in the maths and in the repeat block.
- Putting "change" before "say" in a loop, so the sprite never says the 1st term.

**35–40 min Brain break**

**DO** Pattern moves: clap, clap, stomp, clap, clap, stomp ... Do 12 moves together. Then predict: what is the 20th move? The 30th?

Water. Toilet break for anyone who needs it. Keep it to five minutes.

**40–75 min Worksheet (pairs, then alone, then your choice)**

**DO** Minute 40: give out the workbook. Read the title of Lesson 1 aloud. Say "Pairs first. Talk before you write."

Follow the minute labels on the slides: short teaching slides come before page 1 and before page 2, pairs work together on page 1 (about 40 to 55), each child works alone on page 2 (about 55 to 72), and minutes 72 to 75 are the check together. Any child who has finished can choose Think harder, Go further or the Investigation (all printed in the workbook). The Finish page is for everyone.

**SAY** *"Pairs first. Read each question aloud and talk about it before you write. I am not looking for speed. I am looking for good thinking."*

**DO** Circulate and ask:

- Ask: "What is the difference? Which times table is it like?"
- Ask: "Check your rule with the 1st term AND the 4th term. Does it work for both?"
- Ask: "How many jumps are there from the 1st term to the 20th term?"

**IF STUCK** Hint 1: "Write the times table of the difference under the sequence."

Hint 2: "How far is each term from the times-table number under it?"

Hint 3: "Count the jumps for a small case first: from the 1st term to the 4th term is 3 jumps."

**DO** Think harder, Go further and the Investigation are open to every child, with no label. The Investigation is printed on the Finish page that every child gets. After question 3 (see the line printed on page 1) any child may choose to carry on or try Think harder. Never announce who is on which page. Do not split a pair unless both children agree.

## Stretch and challenge questions

| Question | Answer |
|---|---|
| Workbook Think harder 1: Find the 100th term of 12, 19, 26, 33, … | 705. The rule is 7n + 5, and 7 × 100 + 5 = 705. |
| Workbook Think harder 2: Is 200 a term of 2, 9, 16, 23, …? Explain. | No. The rule is 7n − 5. The 29th term is 198 and the 30th term is 205, so 200 is skipped. |
| Workbook Think harder 3: Matchstick houses in a row: 1 house uses 6 sticks, 2 houses use 11, 3 houses use 16. How many houses can you make with 100 sticks? | 19 houses. The rule is 5n + 1: 19 houses use 96 sticks (4 left over) and 20 houses would need 101. |
| Workbook Think harder 4: Write the difference, the rule for the nth term and the 100th term of (a) 9, 11, 13, 15, … and (b) 0.5, 1, 1.5, 2, … | (a) Difference 2; 2n + 7; 207. (b) Difference 0.5; 0.5n (or n ÷ 2); 50. |
| Workbook Go further 5: Make up your own sequence puzzle. Choose a first term and a difference, write the first four terms, then ask: “Is ___ a term?” Pick a number that is NOT a term but is close to one. Swap with a partner and explain how you know. | Open. Example: 6, 11, 16, 21, … (rule 5n + 1). Is 100 a term? No: 100 − 1 = 99 is not a multiple of 5 (the 19th term is 96 and the 20th is 101). Look for a reason from the rule or from counting jumps, not just a list. |
| Workbook Go further 6: Which numbers below 60 are in BOTH sequences 2, 5, 8, 11, … and 4, 9, 14, 19, …? Find them all. | 14, 29, 44 and 59. One more than each of them is a multiple of both 3 and 5, so a multiple of 15 (15, 30, 45, 60). |
| Workbook Go further 7: What does this code say? set t to 0, set k to 0, then repeat 5 [change k by 1, change t by k], then say t. What would it say with repeat 100? | 15 (1 + 2 + 3 + 4 + 5). With repeat 100 it says 5050: pair 1 + 100, 2 + 99, … to make 50 pairs of 101. |

| | |
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| Workbook Investigation, Finish page (What if we change ...? What stays the same?): Triangles in a row use 3, 5, 7, ... sticks (the rule is 2n + 1). What if we join squares in a row, or pentagons, or hexagons, sharing one side each time? Pick a shape. Draw patterns 1, 2 and 3, make a table and find the rule. What changed? What stayed the same? First step: draw 3 squares in a row and count the sticks. | Regular shapes with k sides joined along one side each time use (k − 1) × n + 1 sticks: triangles 2n + 1, squares 3n + 1, pentagons 4n + 1, hexagons 5n + 1. What changes: the number added each time, which is always one less than the number of sides because a side is shared. What stays the same: the rule has the form (something) × n + 1, and pattern 1 is the single shape (k sticks). Look for: a table, the difference named, a check of the rule on pattern 1 and one more pattern, and a reason for the sharing. Any shape chosen is valid. Stretch: a child may write the rule for any k, or test it in code (set s to k, repeat n − 1 [change s by k − 1]). |

These are the questions on the Think harder, Go further and Finish pages in the workbook. More are in the separate Question Bank.

### 75–82 min Talk time

**DO** Bring the group back to a circle (or camera on, mics on). Ask for volunteers first and choose two or three of them, not only the fastest. Never cold-call a child by name. “Pass” is always fine, and the next person helps. No public tallies, points or rankings.

**SAY** *“Ali says the 100th term of 7, 10, 13, ... is 7 + 3 × 100 = 307. Is he right?”*

*“When is the nth-term rule better than adding on? When is code better than both?”*

| Discussion question | Listen for |
|---|---|
| Ali says the 100th term of 7, 10, 13, ... is 7 + 3 × 100 = 307. Is he right? Explain with a rule, a picture or code. | No. The 1st term is already 7 before any jump, so the 100th term is only 99 jumps away: 7 + 3 × 99 = 304. The rule 3n + 4 gives 304 too. The rule is fastest for one far term; code is best for listing many terms or checking an answer. |

### 82–90 min Exit slip, looking back and closing

**DO** Minutes 82 to 86: exit slip. Children write on the exit slip box of the Finish page (or a slip of paper). No talking, no helping. Show the exit ticket slide.

| Exit slip | Answer |
|---|---|
| The sequence 8, 11, 14, 17, ... (a) Write the rule for the nth term. (b) Find the 100th term. (c) A loop starts at the 1st term. How many times must it repeat "change by 3" to reach the 100th term? | (a) 3n + 5 (b) 305 (c) 99 times |

**SAY** *“Today you found the 100th term without writing 100 numbers, and you tested it in code. That is what mathematicians and programmers both do.”*

*“Do the exit ticket on your own, then show me.”*

## Looking back (2 minutes, minutes 86 to 88)

**SAY** *"Two minutes, just for you, in the "Looking back" box on the Finish page. Finish: something I did well, the hardest part was, and tick what helped you. You can write "pass". If you are still stuck, tick that box and I will check in with you in the open session, in the shared room, with your parent nearby. Nobody marks this and nobody has to show it."*

**DO** Everyone reaches this box: it is on the Finish page, not on the Think harder pages. Collect nothing in front of the group and never read an answer out. You may look at a child's box on camera in the open session if the child offers it. Praise the strategy, not the speed: "you you counted the jumps, and that is what got you unstuck", never "you were quick".

If a child ticks "I am still stuck", check in during the open session the same day: visit the shared breakout room or use a hand signal, with a parent nearby. Never by private message. If a child writes "everything" or "I am bad at maths", say quietly in the open session: "Thank you for telling me. Let's find one small thing that worked today," and point to a specific item they got right. If a child writes "nothing was hard", ask what would make it harder.

**DO** Minutes 88 to 90: stamp the inquiry journal when a child shows you the exit slip. Mark S, D or H in the progress tracker at the back of this guide. If a child marked D or H, plan two minutes of catch-up at the start of the next lesson.

## Extra idea for anyone who wants it

Make the program say the position and the term together (for example "term 7 is 23") using a second variable pos and the join block. Then make it stop at the first term bigger than 1000.

## Answer key (workbook)

**Q1** (a) 4, 7, 10, 13, 16, 19. Each new square adds 3 sticks, not 4, because it shares one side with the square before it. (b) sticks = 3 × n + 1 (c) 301 (d) Pattern 20 (3 × 20 + 1 = 61)

**Q2** (a) difference 4; 4n + 2; 402 (b) 5; 5n − 4; 496 (the other two sequences of this question are now Think harder 4)

**Q3** (a) 4n + 1 (b) Yes. 101 − 1 = 100 = 4 × 25, so 101 is the 25th term. (c) No. Every term is 1 more than a multiple of 4, so every term is odd; 150 is even. (150 − 1 = 149 is not a multiple of 4.)

**Q4** Trace: x is 7, 12, 17, 22. (a) 22 (b) 502. x starts at 2 and 100 jumps of 5 add 500.

**Q5** Program: when green flag clicked; set n to 4; repeat 20 [say n for 1 seconds; change n by 6]. Say must come before change (or start n at −2 and change first). (a) 118 (the 20th term: 6 × 20 − 2), the last number it says. It says one term per second.

**Q6** (a) 203 (3 + 50 × 4) (b) 199 (4 × 50 − 1) (c) Change repeat 50 to repeat 49. The 1st term is already there before the loop starts, so the 50th term is only 49 jumps away.

**Investigation** Regular shapes with k sides joined along one side each time use (k − 1) × n + 1 sticks: triangles 2n + 1, squares 3n + 1, pentagons 4n + 1, hexagons 5n + 1. What changes: the number added each time, which is always one less than the number of sides because a side is shared. What stays the same: the rule has the form (something) × n + 1, and pattern 1 is the single shape (k sticks). Look for: a table, the difference named, a check of the rule on pattern 1 and one more pattern, and a reason for the

sharing. Any shape chosen is valid. Stretch: a child may write the rule for any k, or test it in code (set s to k, repeat n – 1 [change s by k – 1]).

# Lesson 2: Prime Hunters

Week 1 (second lesson) • Primes, Factors and Prime Factorisation • Workbook: Lesson 2

| | |
|---|---|
| **Objective** | Find all the factors of a number using factor pairs, decide whether a number is prime by testing small divisors, write a number as a product of primes with a factor tree, use mod (the remainder), if and a counter in code to count factors, and explore Goldbach pairs and numbers with exactly 3 factors. |
| **Thinking skills** | Systematic listing (factor pairs), testing and ruling out, reasoning about when to stop, tracing code, generalising (squares of primes) |
| **Materials** | Laptop or desktop with Scratch 3 open, a printed 1 to 100 grid (or squared paper), two coloured pencils, mini whiteboard, workbook. Optional: a Python editor for children who already know Python. |
| **Online set-up** | Share the 1 to 100 grid on screen and annotate it while children work on their own printed or hand-drawn grid. Build the prime checker together in your shared Scratch window; the blocks are also on the slide. Block colours on our slides are simplified; in Scratch, find blocks by the category name on the left, not by colour. Children hold their grid up to the camera at minute 22. |
| **Prepare** | Solve the worksheet yourself first. Set out one materials box per pair. Open the slides for this lesson and check the answers in the speaker notes. |

### 0–2 min Opening

**SAY** *"Welcome back, inquirers! Today we hunt primes, the building blocks of every whole number."*

*"People have hunted primes for over 2000 years, and today computers hunt them to keep the internet safe. We hunt by hand first, then teach a computer to do it."*

**DO** Greet each child by name. Show the lesson title slide. Read "Today we will…" aloud.

### 2–5 min Sneak-peek puzzles

**SAY** *"Two sneak-peek puzzles. They are ideas we have not learnt yet, so guessing is welcome and "I do not know" is welcome. Nobody is marked. Write your answers on your own paper. When I say "show", everyone holds their paper to the camera at the same moment."*

| Sneak-peek puzzle | Answer and what to listen for |
|---|---|
| Take 12 counters (or draw 12 dots). Make equal rows, for example 3 rows of 4. Find every way to make equal rows with all 12 counters. | 1 row of 12, 2 rows of 6, 3 rows of 4 (and the same turned round: 4 rows of 3, 6 rows of 2, 12 rows of 1). Counting and drawing is enough. |
| Could you make equal rows (more than 1 row and more than 1 counter in each row) with exactly 31 counters? What do you think? Why? | No. 31 only makes 1 row of 31 or 31 rows of 1, because 31 is prime (its only factors are 1 and 31). Children may guess; accept any reason and say we will find out how to be sure. |

**DO** Puzzle 1 is an access puzzle: any child can try it with objects, a drawing or counting, so everyone gets a first win. Puzzle 2 is a thinking puzzle on the idea of today's lesson.

Papers are shown to the camera by everyone at the same moment, or to you only on camera. Never use private chat and never read wrong answers aloud. Reveal the answers to everyone at minute 5 and say: "Both of these are what we learn today."

Do not record the sneak-peek as a mark. Never announce who got which right, never say who will start on which page, and never use the word "advanced" or "top group".

Invite the WHOLE group, never one child: "When you have finished Start here, you can choose Think harder, Go further or the Investigation." Nobody is singled out. The point printed on page 1 ("After question N you may choose…") is the same for all. Any child can choose it, and any child can go back. Do not split a pair unless both children agree.

### 5–10 min Warm-up

**DO** Factor pair hunt. Six number cards. With a partner, find every factor pair and explain one. One card has only one pair: which one? (5 to 8 minutes. Keep it friendly: no racing or timers, and no points for being first.)

**SAY** *"Factor pair hunt. I show a number. With your partner, find every factor pair: every pair of whole numbers that multiply to make it. Take your time. Be ready to explain one pair you found. There is no race and there are no points."*

*"One card is special. See if you can spot which one."*

**DO** At about minute 8, ask the quick questions on the warm-up slides. Use thumbs or a whisper to partners before taking answers. Stop at minute 10.

| Ask | Expect |
|---|---|
| 17 ÷ 5 = 3 remainder __ | 2 |
| 30 ÷ 6 = 5 remainder __ | 0 |
| 23 ÷ 4 = 5 remainder __ | 3 |
| 100 ÷ 7: what is the remainder? | 2 (7 × 14 = 98) |
| A remainder of 0 means …? | It divides exactly: it is a factor. |

### 10–25 min Explore (in pairs, with objects)

**SAY** *"How many prime numbers do you think there are from 1 to 100? Write a guess. Then we find out with a 2000-year-old method called the sieve."*

**DO** 1. On the 1 to 100 grid, cross out 1. It is not prime.

2. Circle 2. Cross out every other multiple of 2.

3. Circle the next number not crossed out (3) and cross out its multiples. Do the same for 5 and 7.

4. Circle every number left. Count the primes.

**SAY** *"Talk to your partner while you work. I will come round and ask you questions."*

**IF STUCK** Stuck: circle 2, then cross out 4, 6, 8, … every 2nd number. Then circle 3 and cross out every 3rd number.

Crossing out the same number twice: "That is fine. 6 is a multiple of 2 AND of 3."

Finished early: circle 11 and look for a multiple of 11 below 100 that is not crossed out yet. Is there one? Why not?

**DO** Go further (from minute 12, on the explore slide, for any pair that wants it): What if the grid went up to 120? Predict which new numbers after 100 will be prime. Which multiples do you need to cross out now: do you need the multiples of 11? Answer: 101, 103, 107, 109 and 113. You still only need the multiples of 2, 3, 5 and 7, because 11 × 11 = 121 is bigger than 120. Nobody has to wait: a pair that has finished can pick this up.

**DO** At about minute 22, say "Pencils down" and bring the group back together. Take the three questions below, one at a time.

| Ask | Expect |
|---|---|
| How many primes are there from 1 to 100? | 25. |
| Why could we stop after crossing out the multiples of 7? | The next prime is 11, and 11 × 11 = 121 is bigger than 100. Every smaller multiple of 11 (22, 33, … 99) has a smaller factor, so it is already crossed out. |
| Is 1 prime? Is 2 prime? | 1 is not: it has only one factor. 2 is prime (factors 1 and 2), the only even prime. |

### 25–35 min Mini-lesson (you teach, they join in)

**DO** Show the mini-lesson slides. Think aloud as you work. Say what you notice, what you test and how you check. Children copy how you think.

**SAY** *"Factors of 36. I work in pairs: 1 × 36, 2 × 18, 3 × 12, 4 × 9, 6 × 6. The next would be 9 × 4, which I already have, so I stop. 9 factors, because 6 only counts once."*

*"Is 91 prime? It is not even. 9 + 1 = 10, not a multiple of 3. It does not end in 0 or 5. Try 7: 7 × 13 = 91. So 91 is NOT prime. It looks prime, but it is not."*

*"How far do I need to test? 10 × 10 = 100 is bigger than 91, so if 91 had a factor pair, one of the pair would be less than 10. I only test 2, 3, 5 and 7."*

*"In code, n mod d is the remainder when n is divided by d. If it is 0, d is a factor, so I change count by 1. At the end, count = 2 means prime."*

| Worked example on the slides | Answer |
|---|---|
| | |

| | |
|---|---|
| Is 91 prime?. Not even; 9 + 1 = 10 is not a multiple of 3; does not end in 0 or 5. Try 7: 91 ÷ 7 = 13. Stop testing once d × d is bigger than the number (10 × 10 = 100). | Not prime: 91 = 7 × 13. |
| Prime checker in code (n = 15). repeat n: if n mod d = 0 then change count by 1; change d by 1. n mod d is the remainder. 0 means d is a factor. At the end, count = 2 means prime. | Factors 1, 3, 5 and 15: count = 4, not 2, so 15 is not prime. |

**SAY** *"The key idea today is: exactly two factors. A prime has exactly 2 factors: 1 and itself. Test 2, 3, 5, 7, … until d × d is bigger than the number. Say it with me."*

**DO** Show the key idea slide at minutes 33 to 35. The "common mistakes" on that slide are for you, not the children. Watch for them during the worksheet:

- Saying 1 is prime, or that 2 is not prime because it is even.
- Missing factors by not working in pairs, or counting the middle factor of a square number twice (6 in 36).
- Stopping a factor tree at a number that is not prime (72 = 2 × 2 × 18), or reading mod as division (thinking 10 mod 4 is 2.5 instead of the remainder 2).

**35–40 min Brain break**

**DO** Prime or not? Jump for prime, squat for not prime: 7, 9, 13, 21, 2, 1, 31, 27, 51, 53.

Water. Toilet break for anyone who needs it. Keep it to five minutes.

**40–75 min Worksheet (pairs, then alone, then your choice)**

**DO** Minute 40: give out the workbook. Read the title of Lesson 2 aloud. Say "Pairs first. Talk before you write."

Follow the minute labels on the slides: short teaching slides come before page 1 and before page 2, pairs work together on page 1 (about 40 to 55), each child works alone on page 2 (about 55 to 72), and minutes 72 to 75 are the check together. Any child who has finished can choose Think harder, Go further or the Investigation (all printed in the workbook). The Finish page is for everyone.

**SAY** *"Pairs first. Read each question aloud and talk about it before you write. I am not looking for speed. I am looking for good thinking."*

**DO** Circulate and ask:

- Ask: "How do you know you have found every factor?"
- Ask: "Why can you stop testing at 7 for numbers below 100?"
- Ask: "Point to the block that adds 1 to count. When does it run?"

**IF STUCK** Hint 1: "Start with 1 × the number, then try 2, 3, 4 … and stop when the pairs meet."

Hint 2: "Use the quick tests: even? digits add to a multiple of 3? ends in 0 or 5? Then try 7."

Hint 3: “Split the number into any two factors you know, then keep splitting until every branch ends in a prime.”

**DO** Think harder, Go further and the Investigation are open to every child, with no label. The Investigation is printed on the Finish page that every child gets. After question 3 (see the line printed on page 1) any child may choose to carry on or try Think harder. Never announce who is on which page. Do not split a pair unless both children agree.

## Stretch and challenge questions

| Question | Answer |
|---|---|
| Workbook Think harder 1: How many factors does 64 have? List them. | 7: 1, 2, 4, 8, 16, 32, 64 (8 × 8 counts once). |
| Workbook Think harder 2: Find the smallest number that has exactly 6 factors. | 12 (1, 2, 3, 4, 6, 12). Every number below 12 has fewer than 6 factors. |
| Workbook Think harder 3: Write 100 as the sum of two primes in every possible way. | 6 ways: 3 + 97, 11 + 89, 17 + 83, 29 + 71, 41 + 59, 47 + 53. |
| Workbook Think harder 4: Write 36 as the sum of two primes in every way. Then find ALL the numbers from 1 to 50 with exactly 3 factors. What is the next one after 50? | 36 = 5 + 31 = 7 + 29 = 13 + 23 = 17 + 19 (4 ways). Exactly 3 factors: 4, 9, 25, 49 (squares of primes). Next after 50: 121 (11 × 11). |
| Workbook Go further 5: Make up a “Who am I?” number puzzle using factors and primes, for example: “I am between 30 and 60 and I have exactly three factors.” Write 2 or 3 clues so that only ONE number fits. Swap with a partner, then check that no other number fits. | Open. Example: “I am between 30 and 60 and I have exactly 3 factors” fits only 49 (7 × 7). The numbers with exactly 3 factors are the squares of primes: 4, 9, 25, 49, 121, … Check a puzzle by listing every number that fits the clues. |
| Workbook Go further 6: A farmer counts her eggs in 2s, 3s, 4s, 5s and 6s, and each time there is 1 egg left over. When she counts them in 7s, none are left over. What is the smallest number of eggs she could have? | 301. It must be 1 more than a multiple of 60: 61, 121, 181, 241, 301, … The first of these in the 7 times table is 301 = 7 × 43. |
| Workbook Go further 7: 20 lockers are closed. Student 1 opens every locker. Student 2 changes every 2nd locker (open to closed, closed to open). Student 3 changes every 3rd locker, and so on up to student 20. Which lockers are open at the end? | 1, 4, 9 and 16. Locker n is changed once for each factor of n. Only square numbers have an odd number of factors, so only they end up open. |

| | |
|---|---|
| Workbook Investigation, Finish page (Is it always true? Test and explain): Ali says: “If you add two different primes, the answer is never a prime.” Test him. List every pair of different primes below 20 and add them. Is Ali right? What do you notice about the pairs that do make a prime? Explain why. First step: write down all the prime numbers below 20. | Ali is wrong. The primes below 20 are 2, 3, 5, 7, 11, 13, 17, 19, giving 28 pairs. Exactly four sums are prime: 2 + 3 = 5, 2 + 5 = 7, 2 + 11 = 13, 2 + 17 = 19. Every one of them uses the prime 2. Why: two odd primes add to an even number bigger than 2, which has 2 as a factor, so it is not prime. 2 is the only even prime, so only a pair with 2 can give an odd sum, and then the sum is prime only if the other prime plus 2 is also prime (3, 5, 11, 17 work; 7, 13, 19 do not). Look for: a systematic list of all 28 pairs, the spot of the number 2, a reason using odd and even. One counter-example is enough to show “never” is false. |

These are the questions on the Think harder, Go further and Finish pages in the workbook. More are in the separate Question Bank.

### 75–82 min Talk time

**DO** Bring the group back to a circle (or camera on, mics on). Ask for volunteers first and choose two or three of them, not only the fastest. Never cold-call a child by name. “Pass” is always fine, and the next person helps. No public tallies, points or rankings.

**SAY** *“Why does a square number like 36 have an odd number of factors?”*

*“Is there a biggest prime? What do you think?”*

| Discussion question | Listen for |
|---|---|
| Why does a square number like 36 have an odd number of factors, when most numbers have an even number? | Factors come in pairs that multiply to the number. For a square number one pair is a number times itself (6 × 6), and that factor is counted only once, so the total is odd. There is no biggest prime: Euclid showed this about 2300 years ago (a teaser for secondary school). |

### 82–90 min Exit slip, looking back and closing

**DO** Minutes 82 to 86: exit slip. Children write on the exit slip box of the Finish page (or a slip of paper). No talking, no helping. Show the exit ticket slide.

| Exit slip | Answer |
|---|---|
| (a) Is 111 prime? Show your test. (b) Write 66 as a product of primes. (c) Optional: is 133 prime? | (a) No. 1 + 1 + 1 = 3, so 3 divides it: 111 = 3 × 37. (b) 66 = 2 × 3 × 11 (c) No. 133 = 7 × 19. |

**SAY** *“You sieved, you tested, you split numbers into primes, and you taught a computer to count factors. Real prime hunters!”*

*“Do the exit ticket on your own, then show me.”*

## Looking back (2 minutes, minutes 86 to 88)

**SAY** *"Two minutes, just for you, in the "Looking back" box on the Finish page. Finish: something I did well, the hardest part was, and tick what helped you. You can write "pass". If you are still stuck, tick that box and I will check in with you in the open session, in the shared room, with your parent nearby. Nobody marks this and nobody has to show it."*

**DO** Everyone reaches this box: it is on the Finish page, not on the Think harder pages. Collect nothing in front of the group and never read an answer out. You may look at a child's box on camera in the open session if the child offers it. Praise the strategy, not the speed: "you you listed the factors in pairs, and that is what got you unstuck", never "you were quick".

If a child ticks "I am still stuck", check in during the open session the same day: visit the shared breakout room or use a hand signal, with a parent nearby. Never by private message. If a child writes "everything" or "I am bad at maths", say quietly in the open session: "Thank you for telling me. Let's find one small thing that worked today," and point to a specific item they got right. If a child writes "nothing was hard", ask what would make it harder.

**DO** Minutes 88 to 90: stamp the inquiry journal when a child shows you the exit slip. Mark S, D or H in the progress tracker at the back of this guide. If a child marked D or H, plan two minutes of catch-up at the start of the next lesson.

## Extra idea for anyone who wants it

First add if count = 2 then say prime, else say not prime after the loop. Then put the prime checker inside a second loop so n goes 1, 2, 3, … 100 and the sprite says only the primes. Does it find 25 primes, like the sieve?

## Answer key (workbook)

**Q1** 28: 1, 2, 4, 7, 14, 28 (6 factors, not prime). 31: 1, 31 (2, prime). 49: 1, 7, 49 (3, not prime). 51: 1, 3, 17, 51 (4, not prime: 51 = 3 × 17). 1: 1 (1 factor, not prime). (a) 67, 83 and 97 are prime (57 = 3 × 19, 77 = 7 × 11, 99 = 9 × 11). (b) No. 8 + 7 = 15, a multiple of 3, so 87 = 3 × 29.

**Q2** (a) 72 = 2 × 2 × 2 × 3 × 3 (b) 150 = 2 × 3 × 5 × 5 (c) 126 (2 × 3 × 3 × 7)

**Q3** d = 1 to 6. 6 mod d: 0, 0, 0, 2, 1, 0. count after each loop: 1, 2, 3, 3, 3, 4. (a) 4 (the factors 1, 2, 3, 6) (b) It says 2, so 13 is prime: its only factors are 1 and 13.

**Q4** (a) 24 = 5 + 19 = 7 + 17 = 11 + 13 (3 ways) (b) An odd total needs one even and one odd number. The only even prime is 2, and 27 − 2 = 25 is not prime.

**Q5** (a) 4, 9, 25 (three numbers up to 30) (b) They are squares of primes (2 × 2, 3 × 3, 5 × 5). Their factors are 1, the prime and the square. (The 36 and the 50 versions are Think harder 4.)

**Q6** E, C, B. Inside B: F (with A inside F), then D. It says 1, 2, 4, 5, 8, 10, 20, 40 (8 numbers, one per second).

**Investigation** Ali is wrong. The primes below 20 are 2, 3, 5, 7, 11, 13, 17, 19, giving 28 pairs. Exactly four sums are prime: 2 + 3 = 5, 2 + 5 = 7, 2 + 11 = 13, 2 + 17 = 19. Every one of them uses the prime 2. Why: two odd primes add to an even number bigger than 2, which has 2 as a factor, so it is not prime. 2 is the only even prime, so only a pair with 2 can give an odd sum, and then the sum is prime only if the other prime plus 2 is also prime (3, 5, 11, 17 work; 7, 13, 19 do not). Look for: a systematic list of all 28

pairs, the spot of the number 2, a reason using odd and even. One counter-example is enough to show “never” is false.

# Lesson 3: Below Zero

Week 2 (first lesson) • Negative Numbers • Workbook: Lesson 3

| | |
|---|---|
| **Objective** | Order and compare negative numbers on a number line; add and subtract with negative numbers by moving along the line, in temperature, lift and sea-level contexts; use "change by" a negative number in code and trace a variable that goes below 0. |
| **Thinking skills** | Number line reasoning, splitting a jump at 0, difference as a distance, tracing code, debugging, planning a loop |
| **Materials** | Laptop or desktop with Scratch 3 (scratch.mit.edu, free, no log-in needed), mini whiteboard and marker, workbook, pencil. |
| **Online set-up** | Children keep Scratch open in one browser tab and the video call in another. The tutor screen-shares Scratch to show the x number under the stage. Children share their screens only when asked; page 1 of the worksheet is done on paper with laptops turned away. Type the minus sign with the keyboard minus key (the hyphen); do not copy and paste a number from the slide or chat, because the slide uses a different minus character that Scratch will not accept. Block colours on our slides are simplified; in Scratch, find blocks by the category name on the left, not by colour. |
| **Prepare** | Solve the worksheet yourself first. Set out one materials box per pair. Open the slides for this lesson and check the answers in the speaker notes. |

### 0–2 min Opening

**SAY** *"Welcome back, inquirers! In Secondary 1 you will meet numbers that go below zero. You already know some: the freezer, a winter holiday in Japan, the basement car park. Today we find out how they behave, and we make the computer go below zero too."*

*"Notice it, test it, explain why. Today "test it" means test it in Scratch."*

**DO** Greet each child by name. Show the lesson title slide. Read "Today we will…" aloud.

### 2–5 min Sneak-peek puzzles

**SAY** *"Two sneak-peek puzzles. They are ideas we have not learnt yet, so guessing is welcome and "I do not know" is welcome. Nobody is marked. Write your answers on your own paper. When I say "show", everyone holds their paper to the camera at the same moment."*

| Sneak-peek puzzle | Answer and what to listen for |
|---|---|
| It is 3 °C outside. The temperature falls by 5 degrees. Draw a thermometer or a number line. What is the temperature now? | −2 °C (3 down to 0 is 3 degrees, then 2 more below zero). A drawn thermometer or number line shows it. |
| A freezer is at −18 °C and a fridge is at 4 °C. How many degrees warmer is the fridge than the freezer? | 22 degrees. From −18 up to 0 is 18, then 4 more: 18 + 4 = 22. A number line drawing shows it. Guessing is fine. |

**DO** Puzzle 1 is an access puzzle: any child can try it with objects, a drawing or counting, so everyone gets a first win. Puzzle 2 is a thinking puzzle on the idea of today's lesson.

Papers are shown to the camera by everyone at the same moment, or to you only on camera. Never use private chat and never read wrong answers aloud. Reveal the answers to everyone at minute 5 and say: "Both of these are what we learn today."

Do not record the sneak-peek as a mark. Never announce who got which right, never say who will start on which page, and never use the word "advanced" or "top group".

Invite the WHOLE group, never one child: "When you have finished Start here, you can choose Think harder, Go further or the Investigation." Nobody is singled out. The point printed on page 1 ("After question N you may choose…") is the same for all. Any child can choose it, and any child can go back. Do not split a pair unless both children agree.

### 5–10 min Warm-up

**DO** Cold city line-up. Five cities at 7 a.m. on a December morning: Singapore 25 °C, Tokyo 3 °C, Seoul −7 °C, Beijing −9 °C, Harbin −22 °C. Put them in order from coldest to warmest, then find how many degrees warmer Tokyo is than Seoul. (5 to 8 minutes. Keep it friendly: no racing or timers, and no points for being first.)

**SAY** *"Here are five cities at 7 a.m. on a December morning. Put them in order from coldest to warmest. Beijing is −9 and Seoul is −7. Nine is bigger than seven, so which is colder? Argue with your partner."*

*"How many degrees warmer is Tokyo than Seoul? How did you count it?"*

**DO** At about minute 8, ask the quick questions on the warm-up slides. Use thumbs or a whisper to partners before taking answers. Stop at minute 10.

| Ask | Expect |
|---|---|
| It is 3 °C. It falls 5 degrees. What is it now? | −2 °C |
| Which is colder: −7 °C or −2 °C? | −7 °C |
| Count back in 3s: 6, 3, 0, __, __ | −3, −6 |
| What number is 4 less than 1? | −3 |
| A lift is at B1. It goes down 2 floors. Where is it? | B3 |

### 10–25 min Explore (in pairs, with objects)

**SAY** *"Your sprite is at x = 0. It changes x by 100, then changes x by −250. Where will it be? Write your prediction before you test it."*

*"The thermometer starts at 5 and goes down 3, six times. What is the last number it will say? Write it down first."*

**DO** 1. In Scratch, set x to 0: the sprite is in the middle of the stage.

2. Predict, then test: change x by 100, then change x by −250. Where is the sprite?

3. Build the thermometer: set temp to 5, repeat 6: change temp by −3, say temp.

4. Predict the last number it says. Then click the flag and check.

**SAY** *"Talk to your partner while you work. I will come round and ask you questions."*

**IF STUCK** Stuck: drag the sprite left and right and watch the x number under the stage. Left of the middle is negative.

Wrong prediction: draw a number line on your whiteboard and do the six jumps of 3. Where do you cross 0?

Finished early: make the sprite walk with the temperature (set x to temp × 10), or add "if temp < 0 then say Below zero!".

**DO** Go further (from minute 12, on the explore slide, for any pair that wants it): What if the thermometer program starts at 5 and repeats 20 times (change temp by −3 each time)? Predict the last number. Which repeat number first gives a number below −30? Answer: −55 (5 − 3 × 20). Below −30 first happens at repeat 12: 5 − 36 = −31 (repeat 11 gives −28). Nobody has to wait: a pair that has finished can pick this up.

**DO** At about minute 22, say "Pencils down" and bring the group back together. Take the three questions below, one at a time.

| Ask | Expect |
|---|---|
| Where did the sprite end up after +100 and then −250? | x = −150 |
| What was the last number the thermometer said? | −13 (2, −1, −4, −7, −10, −13) |
| How do we make a variable go down in Scratch? | Change it by a negative number: adding a negative moves left. |

**25–35 min Mini-lesson (you teach, they join in)**

**DO** Show the mini-lesson slides. Think aloud as you work. Say what you notice, what you test and how you check. Children copy how you think.

**SAY** *"It is −4 °C and it rises 9 degrees. Rising means moving right. From −4 it takes 4 to get to 0. I still have 5 of my 9 left, so I land on 5 °C."*

*"Change temp by −4 means I add negative 4. Adding a negative moves me left, so it is the same as taking away 4: 1 + (−4) = 1 − 4 = −3."*

*"7 − (−3) asks how far it is from −3 to 7. From −3 up to 0 is 3, from 0 up to 7 is 7. That is 10 steps right, so the answer is 10. Check: −3 + 10 = 7."*

| Worked example on the slides | Answer |
|---|---|
| | |

| | |
|---|---|
| Up and down the line. It is −4 °C at 6 a.m. It rises 9 degrees by noon, then falls 7 degrees by night. Split each jump at 0: −4 up to 0 is 4, then 5 more; 5 down to 0, then 2 more. What is the temperature at noon, and at night? | Noon 5 °C, night −2 °C. |
| Subtracting: the jump from b to a. How much warmer is 7 °C than −3 °C? That is 7 − (−3). Start at −3 and count to 7: 3 up to 0, then 7 more. What is 7 − (−3)? | 10 (check: −3 + 10 = 7). |

**SAY** *"The key idea today is: move along the number line. Right is bigger. Adding a positive moves right; adding a negative moves left. a − b is the jump from b to a. Say it with me."*

**DO** Show the key idea slide at minutes 33 to 35. The "common mistakes" on that slide are for you, not the children. Watch for them during the worksheet:

- Ordering negatives by their digits only (saying −9 is bigger than −4).
- Crossing zero wrongly, e.g. 3 − 8 answered as 5 or −11, instead of splitting the jump at 0 (3 down to 0, then 5 more: −5).
- In code, using "change by 1" when the variable should go down, or counting the start value as one of the repeats when tracing.

**35–40 min Brain break**

**DO** Human lift. Standing tall is floor 0; reach up for each floor up, bend lower for each floor down. Moves: up 3, down 5, down 2, up 6. Call out the floor after each move (3, −2, −4, 2).

Water. Toilet break for anyone who needs it. Keep it to five minutes.

**40–75 min Worksheet (pairs, then alone, then your choice)**

**DO** Minute 40: give out the workbook. Read the title of Lesson 3 aloud. Say "Pairs first. Talk before you write."

Follow the minute labels on the slides: short teaching slides come before page 1 and before page 2, pairs work together on page 1 (about 40 to 55), each child works alone on page 2 (about 55 to 72), and minutes 72 to 75 are the check together. Any child who has finished can choose Think harder, Go further or the Investigation (all printed in the workbook). The Finish page is for everyone.

**SAY** *"Pairs first. Read each question aloud and talk about it before you write. I am not looking for speed. I am looking for good thinking."*

**DO** Circulate and ask:

- Ask: "Which way did you move? Where did you cross zero?"
- Ask: "How can you check your answer by adding back?"
- Ask: "In your trace table, which row is the start and which rows are the repeats?"

**IF STUCK** Hint 1: "Draw a number line and mark 0 first."

Hint 2: “Split the jump at 0: how far to 0, and how far after it?”

Hint 3: “For a − b, start at b and count to a. Right means a positive answer, left means a negative answer.”

**DO** Think harder, Go further and the Investigation are open to every child, with no label. The Investigation is printed on the Finish page that every child gets. After question 3 (see the line printed on page 1) any child may choose to carry on or try Think harder. Never announce who is on which page. Do not split a pair unless both children agree.

## Stretch and challenge questions

| Question | Answer |
|---|---|
| Workbook Think harder 1: Work out −7 + 12 and 5 − 13. | 5 and −8 |
| Workbook Think harder 2: A diver is at −14 m. A seagull is flying at 9 m. How far apart are they? | 23 m (14 up to sea level, then 9 more) |
| Workbook Think harder 3: Work out 4 − (−6). Hint: how far is it from −6 to 4? | 10 |
| Workbook Think harder 4: Find the missing number: −3 + ____ = −10. | −7 (adding −7 moves 7 to the left) |
| Workbook Go further 5: Make up your own temperature story with at least three changes (rises and falls) that starts below zero and ends at exactly −7 °C. Write the story, the working and a number-line sketch. Swap with a partner and check each other’s. | Open. Example: starts at −2 °C, falls 6 degrees (−8), rises 3 degrees (−5), falls 2 degrees (−7). Check that the story really ends at −7 and that each change is shown on a number line. |
| Workbook Go further 6: Two numbers add up to −2. Their difference is 10. What are the two numbers? | 4 and −6 (4 + (−6) = −2 and 4 − (−6) = 10) |
| Workbook Go further 7: In some Singapore buildings there is no level 0. (Q6 on page 2 uses 0 for the ground floor; here, level 1 is the ground floor and B1 is the level below it.) A lift goes from level 4 down to B3. How many floors does it go down? (Count the levels one by one; do not use a number line with 0.) | 6 floors (4, 3, 2, 1, B1, B2, B3: six moves down). A number line with 0 would give 7, because it counts a level 0 that these buildings do not have. |

| | |
|---|---|
| Workbook Investigation, Finish page (Find all the ways): Here are some numbers: −5, −3, −1, 0, 2, 4, 6. Find ALL the pairs (a, b) from the list with a − b = 7. Mark each pair on a number line. What do you notice? Then try again with a − b = 3. First step: draw a number line from −5 to 6. Pick a = 2 and ask: what number b makes 2 − b = 7? | a − b = 7: (2, −5), (4, −3) and (6, −1): exactly three pairs (checked by trying every pair in the list). On the number line each pair is the same distance, 7, apart: the same jump slid along the line. a − b = 3: (0, −3) and (2, −1): two pairs. What stays the same: the gap between a and b. Look for: a systematic search (take each b and ask what a would be, then check a is on the list), a jump drawn on a number line, and the idea that "difference" means distance. Children who say (7, 0) have gone outside the list: ask "is 7 on the list?". |

These are the questions on the Think harder, Go further and Finish pages in the workbook. More are in the separate Question Bank.

### 75–82 min Talk time

**DO** Bring the group back to a circle (or camera on, mics on). Ask for volunteers first and choose two or three of them, not only the fastest. Never cold-call a child by name. "Pass" is always fine, and the next person helps. No public tallies, points or rankings.

**SAY** *"Ben says −8 is bigger than −3 because 8 is bigger than 3. Is he right?"*

*"Why does change by −3 make a number go down?"*

| Discussion question | Listen for |
|---|---|
| Ben says −8 is bigger than −3 because 8 is bigger than 3. Is he right? Use a thermometer or a lift to explain. | Ben is not right. −8 is further left on the number line, so it is smaller: −8 °C is colder than −3 °C and floor −8 is lower than floor −3. The digit tells us how far from 0, the sign tells us which side. Change by −3 adds a negative, which moves left, so it is the same as taking away 3. |

### 82–90 min Exit slip, looking back and closing

**DO** Minutes 82 to 86: exit slip. Children write on the exit slip box of the Finish page (or a slip of paper). No talking, no helping. Show the exit ticket slide.

| Exit slip | Answer |
|---|---|
| It is −3 °C. The temperature falls 6 degrees, then rises 4 degrees. What is the temperature now? | −5 °C (−3 − 6 = −9, then −9 + 4 = −5) |

**SAY** *"Today you went below zero on paper and in code. In Secondary 1 these are called negative numbers or integers, and you already know how to move with them. Well done, inquirers!"*

### Looking back (2 minutes, minutes 86 to 88)

**SAY** *"Two minutes, just for you, in the "Looking back" box on the Finish page. Finish: something I did well, the hardest part was, and tick what helped you. You can write "pass". If you are still stuck, tick that box and I will check in with you in the open session, in the shared room, with your parent nearby. Nobody marks this and nobody has to show it."*

**DO** Everyone reaches this box: it is on the Finish page, not on the Think harder pages. Collect nothing in front of the group and never read an answer out. You may look at a child's box on camera in the open session if the child offers it. Praise the strategy, not the speed: "you you drew a number line, and that is what got you unstuck", never "you were quick".

If a child ticks "I am still stuck", check in during the open session the same day: visit the shared breakout room or use a hand signal, with a parent nearby. Never by private message. If a child writes "everything" or "I am bad at maths", say quietly in the open session: "Thank you for telling me. Let's find one small thing that worked today," and point to a specific item they got right. If a child writes "nothing was hard", ask what would make it harder.

**DO** Minutes 88 to 90: stamp the inquiry journal when a child shows you the exit slip. Mark S, D or H in the progress tracker at the back of this guide. If a child marked D or H, plan two minutes of catch-up at the start of the next lesson.

## Extra idea for anyone who wants it

Make Dot walk along the stage as the temperature changes: inside the loop add "set x to temp × 10" and "if temp < 0 then say Below zero!". Predict where Dot stops before running it.

## Answer key (workbook)

**Q1** (a) A = −8, B = −3, C = 4 (b) −12, −5, −1, 0, 3, 7 (c) −4 > −9; −1 < 0; −15 < 2

**Q2** (a) 3 p.m. (−11 °C) (b) 11 degrees (−24 up to −13) (c) 9 degrees (−11 down to −20) (d) −25 °C (e) 44 degrees (13 up to 0, then 31 more)

**Q3** (a) 15 m (6 up to sea level, then 9 more) (b) 9 m (−6 down to −15) (c) −13 m (−6 + 4 = −2, then −2 − 11 = −13)

**Q4** (a) 3 (b) −7 (c) −8 (d) −8 (e) 8 (from −3 to 5) (f) 6 (from −7 to −1)

**Q5** (a) Table: start 40; 1: 10; 2: −20; 3: −50; 4: −80. The sprite says −80. (b) 4 more repeats (−80 to −200 is 120, and 120 ÷ 30 = 4)

**Q6** (a) 13 (the lift goes up: 5 + 8) (b) change floor by −1 (c) −3, the third basement (B3)

**Q7** (a) set depth to 0; repeat 7; change depth by −4; say depth (7 × 4 = 28) (b) 28 (change depth by 28 brings the diver back to 0 m)

**Investigation** a − b = 7: (2, −5), (4, −3) and (6, −1): exactly three pairs (checked by trying every pair in the list). On the number line each pair is the same distance, 7, apart: the same jump slid along the line. a − b = 3: (0, −3) and (2, −1): two pairs. What stays the same: the gap between a and b. Look for: a systematic search (take each b and ask what a would be, then check a is on the list), a jump drawn on a number line, and the idea that "difference" means distance. Children who say (7, 0) have gone outside the list: ask "is 7 on the list?".

# Lesson 4: Algebra Machines

Week 2 (second lesson) • Expressions and Equations • Workbook: Lesson 4

| | |
|---|---|
| **Objective** | Find the rule of a function machine from a table and write it with a letter (OUT = 3n + 2); see that the order of the steps matters and use brackets; solve simple equations by running the machine backwards and checking forwards; build an input-rule-output program and use a search loop to test an answer. |
| **Thinking skills** | Rule finding from a table, writing expressions with a letter, order of operations and brackets, inverse operations, solving simple equations, tracing loops, explaining a number trick with algebra |
| **Materials** | Laptop or desktop with Scratch 3 (scratch.mit.edu, free), mini whiteboard and marker, workbook, pencil. |
| **Online set-up** | Pairs work in breakout rooms for the explore: one partner shares their screen with the machine running and types in the numbers the other partner calls out. Page 1 of the worksheet is done on paper with laptops turned away. Block colours on our slides are simplified; in Scratch, find blocks by the category name on the left, not by colour. |
| **Prepare** | Solve the worksheet yourself first. Set out one materials box per pair. Open the slides for this lesson and check the answers in the speaker notes. |

### 0–2 min Opening

**SAY** *"Today you are machine builders and machine breakers. Every machine has a secret rule. In Secondary 1 the rule is written with a letter, like 3n + 2, and that is all algebra is: a machine written down."*

*"Notice it, test it, explain why. Today we test our rules in code."*

**DO** Greet each child by name. Show the lesson title slide. Read "Today we will…" aloud.

### 2–5 min Sneak-peek puzzles

**SAY** *"Two sneak-peek puzzles. They are ideas we have not learnt yet, so guessing is welcome and "I do not know" is welcome. Nobody is marked. Write your answers on your own paper. When I say "show", everyone holds their paper to the camera at the same moment."*

| Sneak-peek puzzle | Answer and what to listen for |
|---|---|
| A number machine adds 3 to every number you put in. Put in 4, then 7, then 10. What comes out each time? (Count on with objects or your fingers if you like.) | 7, 10 and 13. Every OUT is 3 more than IN. |
| Another machine multiplies by 3, then adds 10. The OUT is 52. What was the IN? | 14. Undo the + 10 first: 52 − 10 = 42. Then undo the × 3: 42 ÷ 3 = 14. Check: 3 × 14 + 10 = 52. Guessing and checking is fine. |

**DO** Puzzle 1 is an access puzzle: any child can try it with objects, a drawing or counting, so everyone gets a first win. Puzzle 2 is a thinking puzzle on the idea of today's lesson.

Papers are shown to the camera by everyone at the same moment, or to you only on camera. Never use private chat and never read wrong answers aloud. Reveal the answers to everyone at minute 5 and say: "Both of these are what we learn today."

Do not record the sneak-peek as a mark. Never announce who got which right, never say who will start on which page, and never use the word "advanced" or "top group".

Invite the WHOLE group, never one child: "When you have finished Start here, you can choose Think harder, Go further or the Investigation." Nobody is singled out. The point printed on page 1 ("After question N you may choose…") is the same for all. Any child can choose it, and any child can go back. Do not split a pair unless both children agree.

### 5–10 min Warm-up

**DO** Guess my rule. The tutor is a machine with a secret rule. Call out IN numbers, hear the OUT, and guess the rule. Rounds: n + 5, 5n, 2n − 1, 3n + 2. Check each rule on every pair. (5 to 8 minutes. Keep it friendly: no racing or timers, and no points for being first.)

**SAY** *"I am a machine. Call out a number and I will tell you what comes out. You say 3, I say 8. You say 7, I say 12. What is my rule?"*

*"Which numbers did you choose to call out? Which ones helped you most?"*

**DO** At about minute 8, ask the quick questions on the warm-up slides. Use thumbs or a whisper to partners before taking answers. Stop at minute 10.

| Ask | Expect |
|---|---|
| n = 6. What is 4n? | 24 |
| n = 6. What is n + 9? | 15 |
| n = 6. What is 3n − 5? | 13 |
| 2n = 18. What is n? | 9 |
| n = 6. What is (n + 4) × 2? | 20 |

### 10–25 min Explore (in pairs, with objects)

**SAY** *"My machine does times 3, then add 2. If you type 4, what will it say? Write your prediction before we run the code."*

*"If I swap the steps, add 2 first and then times 3, will 4 give the same answer? Thumbs up or down."*

**DO** 1. Build the machine in Scratch: ask IN? and wait, set out to (answer * 3) + 2, say out.

2. Change the rule to your own secret two-step rule.

3. Share your screen. Your partner calls IN numbers and you type them in.

4. Your partner writes your rule with n, using as few IN numbers as possible. Then swap.

**SAY** *"Talk to your partner while you work. I will come round and ask you questions."*

**IF STUCK** Stuck building: put the * block inside the + block, then put "answer" in the first gap of the *.

Stuck cracking a rule: ask for IN 0, 1, 2, 3. How much does OUT go up each time?

Finished early: make a three-step machine, or put ask and say in a forever loop.

**DO** Go further (from minute 12, on the explore slide, for any pair that wants it): What if a machine gives OUT 11 when IN is 3, and OUT 19 when IN is 5? Predict its rule, then try your rule on both pairs. Then find the OUT for IN 10. Answer: 4n − 1. IN goes up by 2 and OUT goes up by 8, so the × part is 4; 4 × 3 = 12 is 1 too big, so − 1. OUT for IN 10 is 39. Nobody has to wait: a pair that has finished can pick this up.

**DO** At about minute 22, say "Pencils down" and bring the group back together. Take the three questions below, one at a time.

| Ask | Expect |
|---|---|
| Which IN numbers were most useful? | Numbers in a row show the × part; IN 0 shows the + part. |
| How did you write the rule with n? | The × part first, then the + or − part, e.g. 3n + 2. |
| The machine × 3, + 2 says 20. How can you find IN without guessing? | Run it backwards: 20 − 2 = 18, 18 ÷ 3 = 6. |

**25–35 min Mini-lesson (you teach, they join in)**

**DO** Show the mini-lesson slides. Think aloud as you work. Say what you notice, what you test and how you check. Children copy how you think.

**SAY** *"IN goes up by 1 and OUT goes up by 4 each time. That is what times 4 does. 4 × 1 is 4 but OUT is 7, so I add 3. Rule: 4n + 3. Check: 4 × 4 + 3 = 19."*

*"Machine A adds 2 then times 5; machine B times 5 then adds 2. For 4: (4 + 2) × 5 = 30 and 4 × 5 + 2 = 22. Same steps, different order, different answer. The brackets show what happens first."*

*"5n − 4 = 31. The machine did times 5, then take away 4. I undo the last step first: 31 + 4 = 35. Then undo times 5: 35 ÷ 5 = 7. Check: 7 × 5 − 4 = 31."*

| Worked example on the slides | Answer |
|---|---|
| From a table to a rule. IN 1, 2, 3, 4 gives OUT 7, 11, 15, 19. OUT goes up 4 each time, so × 4. 4 × 1 = 4, but OUT is 7, so + 3. What is the rule? Check it on IN 4. | OUT = 4n + 3 (4 × 4 + 3 = 19). |

| | |
|---|---|
| Run the machine backwards. A machine does × 5, then − 4. OUT is 31. That is 5n − 4 = 31. Undo the last step first: 31 + 4 = 35. Then undo × 5: 35 ÷ 5. What is n? Check forwards. | n = 7 (7 × 5 − 4 = 31). |

**SAY** *"The key idea today is: a rule is a machine. OUT = 3n + 2 means × 3, then + 2. To find IN from OUT, run the machine backwards: undo the last step first. Say it with me."*

**DO** Show the key idea slide at minutes 33 to 35. The "common mistakes" on that slide are for you, not the children. Watch for them during the worksheet:

- Finding a rule from the first pair only (1 → 6 "add 5") without checking it on the other pairs.
- Undoing the steps in the wrong order, or doing the same step again instead of its opposite.
- Ignoring brackets or the order of operations: reading 4n + 1 as "add 1, then times 4".

**35–40 min Brain break**

**DO** Jumping machine. Rule: double it, then take away 1. The tutor calls IN 2, 3, 4, 5; everyone does that many OUT star jumps (3, 5, 7, 9). Last round: "I did 11 jumps. What was my IN?" (6).

Water. Toilet break for anyone who needs it. Keep it to five minutes.

**40–75 min Worksheet (pairs, then alone, then your choice)**

**DO** Minute 40: give out the workbook. Read the title of Lesson 4 aloud. Say "Pairs first. Talk before you write."

Follow the minute labels on the slides: short teaching slides come before page 1 and before page 2, pairs work together on page 1 (about 40 to 55), each child works alone on page 2 (about 55 to 72), and minutes 72 to 75 are the check together. Any child who has finished can choose Think harder, Go further or the Investigation (all printed in the workbook). The Finish page is for everyone.

**SAY** *"Pairs first. Read each question aloud and talk about it before you write. I am not looking for speed. I am looking for good thinking."*

**DO** Circulate and ask:

- Ask: "Did you check your rule on every pair, not just the first?"
- Ask: "What was the last step? What is its opposite?"
- Ask: "How could the computer check your answer?"

**IF STUCK** Hint 1: "How much does OUT go up when IN goes up by 1? That is the × part."

Hint 2: "Draw the machine boxes. Write the OUT at the end and go backwards."

Hint 3: "Put your answer back into the machine. Do you get the OUT?"

**DO** Think harder, Go further and the Investigation are open to every child, with no label. The Investigation is printed on the Finish page that every child gets. After question 3 (see the line printed on page 1) any child may choose to carry on or try Think harder. Never announce who is on which page. Do not split a pair unless both children agree.

## Stretch and challenge questions

| Question | Answer |
|---|---|
| Workbook Think harder 1: IN 1, 2, 3, 4 gives OUT 1, 7, 13, 19. Find the rule. What is OUT when IN is 20? | OUT = 6n – 5; IN 20 gives 115. |
| Workbook Think harder 2: Solve 9n – 7 = 65. | n = 8 (65 + 7 = 72, 72 ÷ 9 = 8) |
| Workbook Think harder 3: Machine A does × 2, then + 10. Machine B does + 5, then × 2. Put 7 into both. What do you notice? Explain with n. | Both give 24. (n + 5) × 2 = 2n + 10, so they always give the same OUT. |
| Workbook Think harder 4: I think of a number, divide it by 4, then add 6. I get 15. What is my number? | 36 (15 – 6 = 9, 9 × 4 = 36) |
| Workbook Go further 5: Find two different machines of the form "× a number, then + or – a number" that both turn IN 5 into OUT 18. Then find the machine of this form that ALSO turns IN 2 into OUT 6. Is there only one? | Many machines turn 5 into 18: × 3 then + 3, × 2 then + 8, × 1 then + 13, × 4 then – 2, … Only × 4 then – 2 also turns 2 into 6 (4 × 2 – 2 = 6). Two pairs of IN and OUT fix the rule: 5a + b = 18 and 2a + b = 6 give 3a = 12, so a = 4 and b = –2. |
| Workbook Go further 6: Mei is n years old. Her mother is 3 times as old. In 12 years, her mother will be twice as old as Mei. How old is Mei now? | 12 (3n + 12 = 2(n + 12), so n = 12; mother 36; in 12 years 24 and 48) |
| Workbook Go further 7: Ali's secret machine does × a number, then + or – a number. IN 4 gives OUT 9. IN 9 gives OUT 24. What does IN 20 give? Which IN gives OUT 99? | The rule is 3n – 3. IN 20 gives 57. OUT 99 comes from IN 34. |
| Workbook Investigation, Finish page (Find all the ways): A machine does "× a number, then + a number". Use whole numbers: the × number is from 1 to 9 and the + number is from 0 to 9. Find ALL the machines that turn IN 4 into OUT 17. What pattern do you spot in your answers? Then try IN 4 and OUT 21. First step: try × 2. Then 4 × 2 = 8. What must be added to 8 to make 17? | IN 4, OUT 17: × 2 then + 9, × 3 then + 5, × 4 then + 1. (× 1 would need + 13, too big; × 5 would need –3, not allowed.) That is exactly three machines. Pattern: when the × number goes up by 1, the + number goes down by 4, because IN is 4 and the total must stay 17. IN 4, OUT 21: × 3 then + 9, × 4 then + 5, × 5 then + 1: again three machines, again down by 4. Look for: a table of × number and + number, a systematic order (try × 1, × 2, × 3, …), and the explanation of the "down by 4". Stretch: a different IN gives a different step (IN 5 steps by 5). |

These are the questions on the Think harder, Go further and Finish pages in the workbook. More are in the separate Question Bank.

### 75–82 min Talk time

**DO** Bring the group back to a circle (or camera on, mics on). Ask for volunteers first and choose two or three of them, not only the fastest. Never cold-call a child by name. "Pass" is always fine, and

the next person helps. No public tallies, points or rankings.

**SAY** *"To solve 4n + 7 = 43, Ravi tries n = 1, 2, 3 … like the computer. Mei runs the machine backwards. Which way is better, and when?"*

*"Why do we undo the LAST step first?"*

| Discussion question | Listen for |
|---|---|
| To solve 4n + 7 = 43, Ravi tries n = 1, 2, 3 … like the computer. Mei runs the machine backwards. Which way is better, and when? | Both give n = 9. Running backwards is quicker by hand and works for big numbers. Trying every n is easy to code and also works when there is no easy backwards step (like n × n + n = 90), but it is slow by hand and must stop if there is no whole-number answer. We undo the last step first because it was the last thing done, like taking off shoes before socks. |

### 82–90 min Exit slip, looking back and closing

**DO** Minutes 82 to 86: exit slip. Children write on the exit slip box of the Finish page (or a slip of paper). No talking, no helping. Show the exit ticket slide.

| Exit slip | Answer |
|---|---|
| A machine does × 4, then + 5. (a) IN 6: what is OUT? (b) OUT 45: what was IN? | (a) 29 (b) 10 (45 – 5 = 40, 40 ÷ 4 = 10) |

**SAY** *"Today you wrote rules with a letter, ran machines backwards and let the computer search. That is the start of Secondary 1 algebra, and you explained it in your own words. Well done, inquirers!"*

## Looking back (2 minutes, minutes 86 to 88)

**SAY** *"Two minutes, just for you, in the "Looking back" box on the Finish page. Finish: something I did well, the hardest part was, and tick what helped you. You can write "pass". If you are still stuck, tick that box and I will check in with you in the open session, in the shared room, with your parent nearby. Nobody marks this and nobody has to show it."*

**DO** Everyone reaches this box: it is on the Finish page, not on the Think harder pages. Collect nothing in front of the group and never read an answer out. You may look at a child's box on camera in the open session if the child offers it. Praise the strategy, not the speed: "you you tried a number and checked, and that is what got you unstuck", never "you were quick".

If a child ticks "I am still stuck", check in during the open session the same day: visit the shared breakout room or use a hand signal, with a parent nearby. Never by private message. If a child writes "everything" or "I am bad at maths", say quietly in the open session: "Thank you for telling me. Let's find one small thing that worked today," and point to a specific item they got right. If a child writes "nothing was hard", ask what would make it harder.

**DO** Minutes 88 to 90: stamp the inquiry journal when a child shows you the exit slip. Mark S, D or H in the progress tracker at the back of this guide. If a child marked D or H, plan two minutes of

catch-up at the start of the next lesson.

### Extra idea for anyone who wants it

Build the search program in Scratch for 8n − 5 = 91. Then change it to look for n × n + n = 90, which has no easy backwards step. What does it say? (9)

### Answer key (workbook)

**Q1** (a) OUT = n + 3; OUT for 10 is 13 (b) OUT for 10 is 33; rule OUT = 3n + 3 (c) IN is 10; rule OUT = 5n − 2 (d) OUT for 20 is 77; rule OUT = 4n − 3

**Q2** (a) P gives 32, Q gives 23 (b) P: OUT = (n + 3) × 4; Q: OUT = 4n + 3 (c) No: (n + 3) × 4 = 4n + 12, so P is always 9 more than Q.

**Q3** (a) n = 8 (43 + 5 = 48, 48 ÷ 6 = 8) (b) n = 10 (74 − 4 = 70, 70 ÷ 7) (c) n = 12 (45 ÷ 5 = 9, 9 + 3) (d) n = 24 (15 − 9 = 6, 6 × 4)

**Q4** (a) 13 (b) −1 (c) 16 (25 + 7 = 32, 32 ÷ 2 = 16)

**Q5** (a) Table: n = 1, 2, 3, 4; it says 5, 9, 13, 17 (b) 4n + 1 (c) 41

**Q6** (a) set n to 0; repeat until (n * 8) − 5 = 91; change n by 1 (b) 12 (c) 91 + 5 = 96, 96 ÷ 8 = 12 ✓

**Q7** (a) 7 gives 3; 10 gives 3 (b) n + 5, 2n + 10, 2n + 6, n + 3, 3 (c) Halving 2n + 6 gives n + 3; taking away n leaves 3, whatever n is.

**Investigation** IN 4, OUT 17: × 2 then + 9, × 3 then + 5, × 4 then + 1. (× 1 would need + 13, too big; × 5 would need −3, not allowed.) That is exactly three machines. Pattern: when the × number goes up by 1, the + number goes down by 4, because IN is 4 and the total must stay 17. IN 4, OUT 21: × 3 then + 9, × 4 then + 5, × 5 then + 1: again three machines, again down by 4. Look for: a table of × number and + number, a systematic order (try × 1, × 2, × 3, ...), and the explanation of the “down by 4”. Stretch: a different IN gives a different step (IN 5 steps by 5).

# Lesson 5: Every Possible Way

Week 3 (first lesson) • Counting • Workbook: Lesson 5

| | |
|---|---|
| **Objective** | Count every possible case without missing one or counting one twice: list in order, use the multiplication rule for choices made one after another, take care with order and with 0 in digit puzzles, count handshakes and grid routes, and test each count with nested loops in code. |
| **Thinking skills** | Systematic listing, the multiplication rule, avoiding double counting, adding as you go (grid routes), tracing and planning nested loops |
| **Materials** | Laptop or desktop with Scratch 3 (scratch.mit.edu) open, the starter project “Every Order” (one loop that says each tea), mini whiteboards, digit cards 0 to 9, squared paper. |
| **Online set-up** | Children share their screen when they test code. The tutor shares the starter project as an .sb3 file in the chat (children use File, Load from your computer; no account needed), and the three starter blocks are also on the slide. Block colours on our slides are simplified; in Scratch, find blocks by the category name on the left, not by colour. Digit cards can be cut from paper. Breakout rooms in pairs for the Explore. |
| **Prepare** | Solve the worksheet yourself first. Set out one materials box per pair. Open the slides for this lesson and check the answers in the speaker notes. |

### 0–2 min Opening

**SAY** *“Welcome back, inquirers! Today’s question sounds easy: how many ways? But counting every possible way, with none missing and none counted twice, is a skill that mathematicians and programmers use every day.”*

*“By the end of today you will make the computer list every case for you, and you will know when its answer is right.”*

**DO** Greet each child by name. Show the lesson title slide. Read “Today we will…” aloud.

### 2–5 min Sneak-peek puzzles

**SAY** *“Two sneak-peek puzzles. They are ideas we have not learnt yet, so guessing is welcome and “I do not know” is welcome. Nobody is marked. Write your answers on your own paper. When I say “show”, everyone holds their paper to the camera at the same moment.”*

| Sneak-peek puzzle | Answer and what to listen for |
|---|---|
| You have 2 tops (red, blue) and 3 pairs of shorts (black, white, grey). Draw or list every different outfit (1 top and 1 pair of shorts). How many outfits? | 6 outfits: red-black, red-white, red-grey, blue-black, blue-white, blue-grey. A drawing or an organised list works. |

| | |
|---|---|
| 5 chess clubs each play every other club once. How many games is that? Is it 5 × 4? | 10 games. 5 × 4 = 20 counts every game twice (A against B is the same game as B against A), so halve it. Guessing is fine. |

**DO** Puzzle 1 is an access puzzle: any child can try it with objects, a drawing or counting, so everyone gets a first win. Puzzle 2 is a thinking puzzle on the idea of today's lesson.

Papers are shown to the camera by everyone at the same moment, or to you only on camera. Never use private chat and never read wrong answers aloud. Reveal the answers to everyone at minute 5 and say: "Both of these are what we learn today."

Do not record the sneak-peek as a mark. Never announce who got which right, never say who will start on which page, and never use the word "advanced" or "top group".

Invite the WHOLE group, never one child: "When you have finished Start here, you can choose Think harder, Go further or the Investigation." Nobody is singled out. The point printed on page 1 ("After question N you may choose…") is the same for all. Any child can choose it, and any child can go back. Do not split a pair unless both children agree.

**5–10 min Warm-up**

**DO** Count it, don't list it. Six quick counting puzzles (pages 15 to 40, two-digit numbers, multiples of 5, lamp posts, numbers with a 7, even numbers). Pairs say how they counted. No racing and no points. (5 to 8 minutes. Keep it friendly: no racing or timers, and no points for being first.)

**SAY** *"Six puzzles. Do not list them all: find a quick way to count, and be ready to say how you counted. Take your time: there is no race."*

*"First one: a book, pages 15 to 40. How many pages? Careful, it is not 25!"*

**DO** At about minute 8, ask the quick questions on the warm-up slides. Use thumbs or a whisper to partners before taking answers. Stop at minute 10.

| Ask | Expect |
|---|---|
| 3 × 4 × 2 = ? | 24 |
| 1 + 2 + 3 + … + 10 = ? | 55 |
| How many whole numbers from 100 to 999? | 900 |
| 6 × 5 ÷ 2 = ? | 15 |
| repeat 5 { repeat 2 { say hi } }: how many his? | 10 |

**10–25 min Explore (in pairs, with objects)**

**SAY** *"The bubble tea shop has 3 teas and 2 toppings. Before you list them, write your guess: how many different orders? Now the shop adds sizes M and L. Will the number of orders go up by 2, or double, or something else? Write a prediction."*

**DO** 1. A bubble tea shop has 3 teas (milk, green, taro) and 2 toppings (pearls, jelly).

2. On paper, list every order of 1 tea and 1 topping. How do you know you have them all?

3. The shop adds sizes M and L. Predict how many orders now, then check.

4. At the computer: open “Every Order” and put a second loop inside the first so the sprite says every order. (Teas are numbered 1 = milk, 2 = green, 3 = taro; toppings 1 = pearls, 2 = jelly. The sprite says join tea topping, so 21 means green with pearls.)

**SAY** *“Talk to your partner while you work. I will come round and ask you questions.”*

**IF STUCK** Stuck listing: write the first tea, then every topping with it. Then the next tea.

Code stuck: drag a second repeat block INSIDE the first one. Which loop should say the topping?

Finished early: add sugar levels 0%, 50% and 100%. Predict the number of orders first, then add a third loop to test it.

**DO** Go further (from minute 12, on the explore slide, for any pair that wants it): What if the shop had 3 teas, 3 sizes (S, M, L) and 3 toppings? Predict how many different orders. Then write how you would check with a list or a loop. Answer: 27 orders (3 × 3 × 3). Fix the tea, then the size, then the topping: three loops inside each other, or a tree with three layers. Nobody has to wait: a pair that has finished can pick this up.

**DO** At about minute 22, say “Pencils down” and bring the group back together. Take the three questions below, one at a time.

| Ask | Expect |
|---|---|
| How many orders with 3 teas and 2 toppings? With sizes M and L too? | 6 orders; with sizes, 12 (each of the 6 orders comes in 2 sizes: 6 × 2). |
| How did you make sure you had every order, with none twice? | Fix the tea, go through every topping, then move to the next tea: a list in order, like a tree. |
| In your code, which loop goes round faster? | The inner loop: it runs all the way round (2 toppings) for every single turn of the outer loop (each tea). |

### 25–35 min Mini-lesson (you teach, they join in)

**DO** Show the mini-lesson slides. Think aloud as you work. Say what you notice, what you test and how you check. Children copy how you think.

**SAY** *“3 shirts and 2 pairs of shorts. I fix the red shirt and try every pair of shorts: red with black, red with grey. That is 2. Every shirt gets the same 2, so 3 shirts give 3 groups of 2: 3 × 2 = 6. Add 2 caps: each of the 6 outfits splits into 2, so 12.”*

*“In code, the outer loop picks the shirt and the inner loop picks the shorts. The inner loop goes all the way round for EACH shirt, so the say block runs 3 × 2 = 6 times.”*

*“5 friends shaking hands. Friend A shakes 4 hands. B has already shaken A’s, so 3 new ones. C: 2 new. D: 1 new. E: none left. 4 + 3 + 2 + 1 = 10. Or: 5 people times 4 hands is 20, but each handshake has two people in it, so I counted every one twice: 20 ÷ 2 = 10.”*

| Worked example on the slides | Answer |
|---|---|
| The multiplication rule: 3 shirts and 2 pairs of shorts. Fix a shirt, then try every pair of shorts: 2 outfits for each shirt. 3 shirts give 3 groups of 2: 3 × 2. Add 2 caps: each outfit splits into 2. | 6 outfits; with caps, 6 × 2 = 12. |
| Handshakes: 5 friends, every pair shakes once. A shakes 4 new hands, B 3 new, C 2, D 1, E 0. Or: 5 × 4 = 20 counts every handshake twice. In code: check every pair (a, b) and keep only a < b. | 4 + 3 + 2 + 1 = 10, or 20 ÷ 2 = 10 handshakes. |

**SAY** *"The key idea today is: every case, exactly once. Work in order: fix the first choice and run through every second choice. Choices made one after another multiply. Say it with me."*

**DO** Show the key idea slide at minutes 33 to 35. The "common mistakes" on that slide are for you, not the children. Watch for them during the worksheet:

- Adding the choices instead of multiplying them (4 T-shirts and 3 shorts gives 7 outfits).
- Counting each handshake twice (7 × 6 = 42) or letting a person shake their own hand.
- Digit cards: putting 0 at the front of a number, or using the same card twice.

**35–40 min Brain break**

**DO** Every possible pose: arms up or arms out (2), and standing, squatting or on one leg (3). That is 2 × 3 = 6 poses. The tutor calls a number from 1 to 6 and everyone does that pose.

Water. Toilet break for anyone who needs it. Keep it to five minutes.

**40–75 min Worksheet (pairs, then alone, then your choice)**

**DO** Minute 40: give out the workbook. Read the title of Lesson 5 aloud. Say "Pairs first. Talk before you write."

Follow the minute labels on the slides: short teaching slides come before page 1 and before page 2, pairs work together on page 1 (about 40 to 55), each child works alone on page 2 (about 55 to 72), and minutes 72 to 75 are the check together. Any child who has finished can choose Think harder, Go further or the Investigation (all printed in the workbook). The Finish page is for everyone.

**SAY** *"Pairs first. Read each question aloud and talk about it before you write. I am not looking for speed. I am looking for good thinking."*

**DO** Circulate and ask:

- Ask: "Which choice did you fix first? How do you know you have them all?"
- Ask: "Point to the inner loop. How many times does it run for one turn of the outer loop?"
- Ask: "Have you counted any pair twice? Has anyone shaken their own hand?"

**IF STUCK** Hint 1: "Fix the first choice, then run through every second choice."

Hint 2: "Draw a tree, or a table with a row for each first choice."

Hint 3: “If the order does not matter, has your method counted each pair twice?”

**DO** Think harder, Go further and the Investigation are open to every child, with no label. The Investigation is printed on the Finish page that every child gets. After question 3 (see the line printed on page 1) any child may choose to carry on or try Think harder. Never announce who is on which page. Do not split a pair unless both children agree.

## Stretch and challenge questions

| Question | Answer |
|---|---|
| Workbook Think harder 1: A set lunch is 1 main, 1 drink and 1 dessert. There are 4 mains and 3 drinks. There are 60 different set lunches. How many desserts are there? | 5 (4 × 3 = 12 and 60 ÷ 12 = 5) |
| Workbook Think harder 2: How many 3-digit numbers can you make from the cards 1, 2, 3 and 4, using each card at most once? How many of them are even? | 24 (4 × 3 × 2); 12 are even (the last card is 2 or 4: 2 choices, then 3 × 2 for the others) |
| Workbook Think harder 3: In a league of 6 teams, every team plays every other team twice (once at home, once away). How many matches are played? | 30 (6 × 5: each ordered pair is one home match) |
| Workbook Think harder 4: (a) Mei packs 4 T-shirts, 3 pairs of shorts and 2 caps. She wants at least 40 outfits with a cap. What is the fewest number of extra pairs of shorts she must buy? (b) A party has 45 handshakes. How many people are there? | (a) 2 extra pairs: with 5 pairs, 4 × 5 × 2 = 40; with 4 pairs only 4 × 4 × 2 = 32. (b) 10 people (9 + 8 + … + 1 = 45, or 10 × 9 ÷ 2 = 45). |
| Workbook Go further 5: Make up your own “how many ways” puzzle whose answer is exactly 24. Write the puzzle, show how you counted, and check with a list or a tree. Swap with a partner. | Open. Examples: 4 children in a line (4 × 3 × 2 × 1 = 24); 2 shirts × 3 shorts × 4 caps = 24 outfits; 3-digit numbers from the cards 1, 2, 3, 4 using each card at most once (4 × 3 × 2 = 24). Check that every case is counted once. |
| Workbook Go further 6: A set of dominoes has one domino for every pair of spot numbers from 0 to 6, including doubles such as 3 \| 3. How many dominoes are in the set? | 28 (7 doubles, plus 6 + 5 + 4 + 3 + 2 + 1 = 21 with two different numbers) |
| Workbook Go further 7: How many 3-digit numbers have digits that add up to 5? (For example, 203.) | 15 (first digit 1: 5 numbers, 2: 4, 3: 3, 4: 2, 5: 1; they are 104, 113, 122, 131, 140, 203, 212, 221, 230, 302, 311, 320, 401, 410, 500) |

| | |
|---|---|
| Workbook Investigation, Finish page (Find all the ways): Pay exactly 50 cents using only 5-cent, 10-cent and 20-cent coins (as many of each as you like). Find ALL the ways. Show how you made sure you did not miss one or count one twice. What if there is also a 50-cent coin? First step: list every way to pay 50 cents that uses two 20-cent coins. | There are 12 ways. Sort by the number of 20-cent coins. Two 20c: the other 10c is 10c or 5c + 5c: 2 ways. One 20c: the other 30c with 10c and 5c coins: 3×10, 2×10 + 2×5, 1×10 + 4×5, 6×5: 4 ways. No 20c: 50c with 10c and 5c coins: 5×10, 4×10 + 2×5, 3×10 + 4×5, 2×10 + 6×5, 1×10 + 8×5, 10×5: 6 ways. Total 2 + 4 + 6 = 12 (checked by brute force). With a 50c coin there is one more way, 13. Look for: an organised list (fix the biggest coin first), a table, no repeats, and a check. Do not accept a list that is only "some ways". |

These are the questions on the Think harder, Go further and Finish pages in the workbook. More are in the separate Question Bank.

### 75–82 min Talk time

**DO** Bring the group back to a circle (or camera on, mics on). Ask for volunteers first and choose two or three of them, not only the fastest. Never cold-call a child by name. "Pass" is always fine, and the next person helps. No public tallies, points or rankings.

**SAY** *"Why does the multiplication rule work for outfits, but 5 × 5 gives the wrong answer for handshakes between 5 friends?"*

*"How would you change a program that lists every pair so that it lists each handshake once?"*

| Discussion question | Listen for |
|---|---|
| Why does the multiplication rule work for outfits, but 5 × 5 gives the wrong answer for handshakes between 5 friends? | For outfits, every shirt goes with every pair of shorts and a shirt-and-shorts outfit is different from any other, so 3 × 2 counts each outfit once. For handshakes, 5 × 5 = 25 includes each person with themselves (5 cases) and counts A with B and B with A as two, so 25 − 5 = 20 and 20 ÷ 2 = 10. In code, keep only the pairs with a < b. |

### 82–90 min Exit slip, looking back and closing

**DO** Minutes 82 to 86: exit slip. Children write on the exit slip box of the Finish page (or a slip of paper). No talking, no helping. Show the exit ticket slide.

| Exit slip | Answer |
|---|---|
| Six friends meet. Every pair of friends shakes hands once. How many handshakes are there? Show how you know. | 15 (5 + 4 + 3 + 2 + 1, or 6 × 5 ÷ 2). |

**SAY** *"Today you counted things that would take ages to list: you fixed one choice, multiplied, and halved when you counted twice. Then the computer listed every case to test you. Notice it, test it, explain why."*

## Looking back (2 minutes, minutes 86 to 88)

**SAY** *"Two minutes, just for you, in the "Looking back" box on the Finish page. Finish: something I did well, the hardest part was, and tick what helped you. You can write "pass". If you are still stuck, tick that box and I will check in with you in the open session, in the shared room, with your parent nearby. Nobody marks this and nobody has to show it."*

**DO** Everyone reaches this box: it is on the Finish page, not on the Think harder pages. Collect nothing in front of the group and never read an answer out. You may look at a child's box on camera in the open session if the child offers it. Praise the strategy, not the speed: "you you listed them in order, and that is what got you unstuck", never "you were quick".

If a child ticks "I am still stuck", check in during the open session the same day: visit the shared breakout room or use a hand signal, with a parent nearby. Never by private message. If a child writes "everything" or "I am bad at maths", say quietly in the open session: "Thank you for telling me. Let's find one small thing that worked today," and point to a specific item they got right. If a child writes "nothing was hard", ask what would make it harder.

**DO** Minutes 88 to 90: stamp the inquiry journal when a child shows you the exit slip. Mark S, D or H in the progress tracker at the back of this guide. If a child marked D or H, plan two minutes of catch-up at the start of the next lesson.

## Extra idea for anyone who wants it

Write a program that counts every 3-digit number whose digits add up to 5 (three loops, one inside another). Then change 5 to other totals: which total gives the most numbers?

## Answer key (workbook)

**Q1** (a) 12 (4 × 3) (b) 24 (4 × 3 × 2) (the "at least 40 outfits" part is now in Think harder 4)

**Q2** (a) 9 (3 choices for the tens, as 0 cannot go first, then 3 for the ones: 3 × 3): 30, 35, 38, 50, 53, 58, 80, 83, 85 (b) 18 (3 × 3 × 2)

**Q3** (a) a 1, b 1: 11; 1, 2: 12; 1, 3: 13; 1, 4: 14; 2, 1: 21; 2, 2: 22 (b) 12 (3 × 4) (c) 34 (d) 20 (5 × 4)

**Q4** (a) 21 (6 + 5 + 4 + 3 + 2 + 1) (b) Every handshake has been counted twice, once for each person: 42 ÷ 2 = 21. (the "45 handshakes" part is now in Think harder 4)

**Q5** (a) 20 (corner numbers, bottom row up: 1 1 1 1 / 1 2 3 4 / 1 3 6 10 / 1 4 10 20) (b) 8 (put 0 at P: 1 1 1 1 / 1 0 1 2 / 1 1 2 4 / 1 2 4 8). 12 routes go through P.

**Q6** Gaps: repeat 7, repeat 7, if a < b (a > b also works: either keeps each pair once). (a) 21, the same as Q4(a); the "if" is checked 7 × 7 = 49 times. (b) It says 7: it counts each person with themselves (a = b), which is not a handshake.

**Investigation** There are 12 ways. Sort by the number of 20-cent coins. Two 20c: the other 10c is 10c or 5c + 5c: 2 ways. One 20c: the other 30c with 10c and 5c coins: 3×10, 2×10 + 2×5, 1×10 + 4×5, 6×5: 4 ways. No 20c: 50c with 10c and 5c coins: 5×10, 4×10 + 2×5, 3×10 + 4×5, 2×10 + 6×5, 1×10 + 8×5, 10×5: 6 ways. Total 2 + 4 + 6 = 12 (checked by brute force). With a 50c coin there is one more way, 13. Look for: an organised list (fix the biggest coin first), a table, no repeats, and a check. Do not accept a list that is only "some ways".

# Lesson 6: Fair or Unfair?

Week 3 (second lesson) • Probability • Workbook: Lesson 6

| | |
|---|---|
| **Objective** | Write a probability as a fraction (ways it can happen out of equally likely outcomes), list all 36 outcomes for two dice, decide whether a game is fair, change a game to make it fair, and run a game 1000 times in code, explaining why the result is close to what we expect but not exact. |
| **Thinking skills** | Listing equally likely outcomes, fractions in simplest form, expected number = fraction × number of tries, judging fairness, simulating with random numbers |
| **Materials** | Laptop or desktop with Scratch 3 (scratch.mit.edu), two dice per pair (or the tutor's dice program), a coin, coloured counters and a paper bag, mini whiteboards, tally sheet. |
| **Online set-up** | Children roll real dice at home; anyone without dice builds the dice program live in the shared Scratch window (pick random 1 to 6, twice) and uses that. This lesson is enrichment: probability is new to most children, so keep it playful. Block colours on our slides are simplified; in Scratch, find blocks by the category name on the left, not by colour. Pairs play the Explore game in breakout rooms and type their tallies into the chat so the group total can be added up. |
| **Prepare** | Solve the worksheet yourself first. Set out one materials box per pair. Open the slides for this lesson and check the answers in the speaker notes. |

### 0–2 min Opening

**SAY** *"Welcome, inquirers! Today we play games and then ask the big question: is this game fair? Our feelings are not enough. We will count outcomes with fractions, and then make the computer play a game a thousand times to test our maths."*

**DO** Greet each child by name. Show the lesson title slide. Read "Today we will…" aloud.

### 2–5 min Sneak-peek puzzles

**SAY** *"Two sneak-peek puzzles. They are ideas we have not learnt yet, so guessing is welcome and "I do not know" is welcome. Nobody is marked. Write your answers on your own paper. When I say "show", everyone holds their paper to the camera at the same moment."*

| Sneak-peek puzzle | Answer and what to listen for |
|---|---|
| A bag has 3 red counters and 5 blue counters. Draw the bag. Without looking, you pick one counter. Which colour is more likely? Why? | Blue, because there are more blue counters (5 out of 8) than red (3 out of 8). Drawing the counters is enough. |
| Flip two coins. Ali wins if both coins match (two heads or two tails). Mei wins if they are different. Do you think the game is fair? List every way the two coins can land. | Fair. The four outcomes are HH, HT, TH, TT. Match: HH and TT (2 out of 4). Different: HT and TH (2 out of 4). A guess is welcome; the list decides. |

**DO** Puzzle 1 is an access puzzle: any child can try it with objects, a drawing or counting, so everyone gets a first win. Puzzle 2 is a thinking puzzle on the idea of today's lesson.

Papers are shown to the camera by everyone at the same moment, or to you only on camera. Never use private chat and never read wrong answers aloud. Reveal the answers to everyone at minute 5 and say: "Both of these are what we learn today."

Do not record the sneak-peek as a mark. Never announce who got which right, never say who will start on which page, and never use the word "advanced" or "top group".

Invite the WHOLE group, never one child: "When you have finished Start here, you can choose Think harder, Go further or the Investigation." Nobody is singled out. The point printed on page 1 ("After question N you may choose…") is the same for all. Any child can choose it, and any child can go back. Do not split a pair unless both children agree.

**5–10 min Warm-up**

**DO** Where on the line? Pairs place six events on a 0 to 1 probability line and give each one's fraction: roll a 7, flip heads, roll less than 5, pick a vowel from SINGAPORE, roll 1 to 6, pick the 1 red card from 10. (5 to 8 minutes. Keep it friendly: no racing or timers, and no points for being first.)

**SAY** *"Here is a line from 0 to 1. 0 means impossible, 1 means certain, 1/2 means an even chance."*

*"For each card, decide where it goes on the line and give its fraction. Take as long as you need, and point to the line first. There is no timer."*

**DO** At about minute 8, ask the quick questions on the warm-up slides. Use thumbs or a whisper to partners before taking answers. Stop at minute 10.

| Ask | Expect |
|---|---|
| Simplify 6/36 | 1/6 |
| 1/6 of 600 | 100 |
| 2/5 of 1000 | 400 |
| Which is bigger: 5/12 or 4/9? | 4/9 (16/36 is more than 15/36) |
| 1 – 3/10 | 7/10 |

**10–25 min Explore (in pairs, with objects)**

**SAY** *"In the Dice Race, player A wins on six totals: 2, 3, 4, 10, 11, 12. Player B wins on only five: 5, 6, 7, 8, 9. Who do you think will win more often? Write A, B or "same", and a reason."*

**DO** 1. Roll two dice and add. A wins on 2, 3, 4, 10, 11 or 12. B wins on 5, 6, 7, 8 or 9.

2. Predict: who will win more often? A has 6 totals, B has only 5.

3. Play 20 rounds with your partner. Tally the wins.

4. Add up the whole group's results. What do you notice?

**SAY** *"Talk to your partner while you work. I will come round and ask you questions."*

**IF STUCK** Keep a tally: one column for A, one for B. Write the total each time.

Surprised by your results? Write down which totals came up most.

Finished early: in how many ways can two dice make a total of 7? Of 2? Use a red die and a blue die so 1 + 6 and 6 + 1 are different.

**DO** Go further (from minute 12, on the explore slide, for any pair that wants it): What if A wins on totals 2 to 6 and B wins on totals 7 to 12? Predict who wins more often. Play 10 rounds, then explain using the 36 outcomes. Answer: B is more likely: totals 7 to 12 can be made in 21 of the 36 ways, totals 2 to 6 in only 15 (so 21/36 = 7/12 against 15/36 = 5/12). The game is not fair. Nobody has to wait: a pair that has finished can pick this up.

**DO** At about minute 22, say "Pencils down" and bring the group back together. Take the three questions below, one at a time.

| Ask | Expect |
|---|---|
| Who won more often in your pair? In the whole group? | Usually B. The group total is closer to B winning about 2 rounds in every 3. |
| A has more totals. Why does B win more often? | The totals are not equally likely: 7 can be made in 6 ways, 2 in only 1 way. B's totals can be made in 24 of the 36 ways. |
| How could we be sure without playing forever? | List all 36 equally likely outcomes, or let a computer play 1000 games. |

### 25–35 min Mini-lesson (you teach, they join in)

**DO** Show the mini-lesson slides. Think aloud as you work. Say what you notice, what you test and how you check. Children copy how you think.

**SAY** *"Roll one dice. More than 4 means 5 or 6: that is 2 ways out of 6 equally likely outcomes. So the probability is 2/6, which is 1/3."*

*"Two dice. The red die has 6 outcomes and for each one the blue die has 6, so 6 × 6 = 36, just like last lesson's nested loop. A total of 7 happens 6 times: 1 + 6, 2 + 5, 3 + 4, 4 + 3, 5 + 2, 6 + 1. So P(7) = 6/36 = 1/6. A total of 2 happens once: 1/36."*

*"In 1000 games I expect B to win about 2/3 × 1000, which is 666 2/3. A count cannot be 666 2/3, and every run is random, so the program will say something close, like 670 or 649, but almost never exactly 667."*

| Worked example on the slides | Answer |
|---|---|
| A spinner with 6 equal parts: 3 blue, 2 yellow, 1 red. 6 equal parts, so 6 equally likely outcomes. Yellow: 2 parts out of 6. Spin 600 times: expect 1/3 of 600. | P(yellow) = 2/6 = 1/3; about 200 yellows in 600 spins (not exactly). |

| | |
|---|---|
| Two dice: the 36 outcomes. 6 outcomes for the red die × 6 for the blue = 36. Total 7: 1+6, 2+5, 3+4, 4+3, 5+2, 6+1. Explore game: A's totals happen 12 ways, B's 24 ways. | P(7) = 6/36 = 1/6. P(A wins) = 12/36 = 1/3, P(B wins) = 24/36 = 2/3: not fair. |

**SAY** *"The key idea today is: count the equally likely outcomes. Probability = ways it can happen ÷ equally likely outcomes. A game is fair when every player has the same probability of winning. Say it with me."*

**DO** Show the key idea slide at minutes 33 to 35. The "common mistakes" on that slide are for you, not the children. Watch for them during the worksheet:

- Treating outcomes as equally likely when they are not (two coins: "two heads, two tails, one of each" as thirds; dice totals 2 to 12 as elevenths).
- Thinking a random result should come out exactly as expected, or that a coin "is due" a head after several tails.
- Not simplifying, or writing the number of ways instead of the fraction (6 instead of 6/36 = 1/6).

**35–40 min Brain break**

**DO** Human dice. The tutor rolls a dice: 1 or 2, five star jumps; 3 or 4, touch your toes five times; 5 or 6, spin round once. Which will we do most? (Each has the same chance: 2/6 = 1/3.)

Water. Toilet break for anyone who needs it. Keep it to five minutes.

**40–75 min Worksheet (pairs, then alone, then your choice)**

**DO** Minute 40: give out the workbook. Read the title of Lesson 6 aloud. Say "Pairs first. Talk before you write."

Follow the minute labels on the slides: short teaching slides come before page 1 and before page 2, pairs work together on page 1 (about 40 to 55), each child works alone on page 2 (about 55 to 72), and minutes 72 to 75 are the check together. Any child who has finished can choose Think harder, Go further or the Investigation (all printed in the workbook). The Finish page is for everyone.

**SAY** *"Pairs first. Read each question aloud and talk about it before you write. I am not looking for speed. I am looking for good thinking."*

**DO** Circulate and ask:

- Ask: "What are the equally likely outcomes here? How many are there?"
- Ask: "What fraction do you expect? So about how many out of 1000?"
- Ask: "Your program said 158, not 167. Is it broken? How could you check?" (run it again; run 10 000 games and look at the fraction)

**IF STUCK** Hint 1: "List every outcome. Are they all equally likely?"

Hint 2: "For two dice, draw the 6 by 6 table."

Hint 3: "Expected number = probability × number of tries."

**DO** Think harder, Go further and the Investigation are open to every child, with no label. The Investigation is printed on the Finish page that every child gets. After question 3 (see the line printed on page 1) any child may choose to carry on or try Think harder. Never announce who is on which page. Do not split a pair unless both children agree.

## Stretch and challenge questions

| Question | Answer |
|---|---|
| Workbook Think harder 1: A spinner has 10 equal parts numbered 1 to 10. What is the probability of spinning a prime number? | 4/10 = 2/5 (2, 3, 5, 7) |
| Workbook Think harder 2: Roll two dice and add. What is the probability that the total is 9? | 4/36 = 1/9 (3+6, 4+5, 5+4, 6+3) |
| Workbook Think harder 3: A bag has 5 red counters and some blue counters. The probability of picking red is 1/4. How many blue counters are there? | 15 (5 out of 20 is 1/4) |
| Workbook Think harder 4: (a) A bag has 4 red and 6 blue counters. How many blue counters must be added so that P(red) = 1/5? (b) In the Difference Game (Ali wins on 0, 1, 2; Ben on 3, 4, 5), change who wins on which differences so that the game is fair. | (a) 10 blue (4 red out of 20: 4/20 = 1/5). (b) Any split with 18 outcomes each, e.g. Ali on 1, 3, 5 (odd) and Ben on 0, 2, 4 (even); or Ali on 1, 2 and Ben on 0, 3, 4, 5; or Ali on 0, 1, 5 and Ben on 2, 3, 4. |
| Workbook Go further 5: Design a FAIR game for two players using one dice, two coins or a spinner. Write the rules and show with a list or table that each player has the same chance of winning. Then change ONE rule to make the game unfair, and say who is favoured. | Open. Example: roll one dice; Player A wins on 1, 2 or 3 and Player B wins on 4, 5 or 6 (3/6 each). Change: A also wins on 4. Now A has 4/6 and B has 2/6, so A is favoured. Check that every outcome is listed and the winning outcomes add up to the whole. |
| Workbook Go further 6: Mei and Ali each roll one dice. Mei wins if her number is bigger. Ali wins if his number is bigger or the same. Is the game fair? What is the probability that Mei wins? | Not fair. Mei wins 15/36 = 5/12; Ali wins 21/36 = 7/12 (15 bigger + 6 the same). Fair version: the same number means roll again. |
| Workbook Go further 7: Spinner A has 3 equal parts: 1, 2, 3. Spinner B has 4 equal parts: 1, 2, 3, 4. Spin both and multiply. Even wins if the answer is even, Odd wins if it is odd. Is the game fair? | Not fair. There are 3 × 4 = 12 outcomes; the answer is odd only when both are odd: 2 × 2 = 4 ways. P(Odd) = 4/12 = 1/3, P(Even) = 8/12 = 2/3. |

| | |
|---|---|
| Workbook Investigation, Finish page (Is it always true? Test and explain): Ali says: “A game with only two outcomes is always fair.” Test him on three games. (1) Flip a coin: heads or tails. (2) Roll a dice: six or not six. (3) Roll two dice: total 7 or not 7. Which games are fair? Is Ali right? Change one unfair game to make it fair. First step: flip a coin 10 times and tally heads and tails. Is a coin fair? Say why. | Ali is wrong. (1) Coin: 1/2 and 1/2, fair. (2) Dice: six has 1/6, not six has 5/6, unfair. (3) Two dice: a total of 7 has 6 out of 36 = 1/6, not 7 has 30 out of 36 = 5/6, unfair. Two outcomes is not the same as two EQUAL chances. Fair means every player has the same probability. Fixes: dice, 1 to 3 against 4 to 6; two dice, even total against odd total (18 out of 36 each, checked by listing all 36). Look for: a list or table of outcomes for game 3, probabilities as fractions, and the idea that one counter-example (game 2) is enough. |

These are the questions on the Think harder, Go further and Finish pages in the workbook. More are in the separate Question Bank.

### 75–82 min Talk time

**DO** Bring the group back to a circle (or camera on, mics on). Ask for volunteers first and choose two or three of them, not only the fastest. Never cold-call a child by name. “Pass” is always fine, and the next person helps. No public tallies, points or rankings.

**SAY** *“Mei flipped a coin 10 times and got 7 heads. She says the coin is unfair. Do you agree?”*

*“How many flips would you need before you believed her? How could a program help?”*

| Discussion question | Listen for |
|---|---|
| Mei flipped a coin 10 times and got 7 heads. She says the coin is unfair. Do you agree? How could you test it? | Not yet: 10 flips is far too few. With a fair coin, 7 or more heads in 10 flips happens 176 times in 1024, about 1 time in 6. To test it, flip many more times (or simulate 1000 flips of a fair coin to see how much the counts wobble). If 1000 real flips gave 700 heads, that would be strong evidence: a fair coin almost never does that. |

### 82–90 min Exit slip, looking back and closing

**DO** Minutes 82 to 86: exit slip. Children write on the exit slip box of the Finish page (or a slip of paper). No talking, no helping. Show the exit ticket slide.

| Exit slip | Answer |
|---|---|
| Roll two dice and add. What is the probability that the total is 8? If you roll them 360 times, about how many 8s do you expect? | 5/36 (2+6, 3+5, 4+4, 5+3, 6+2); about 50 (5/36 × 360 = 50). |

**SAY** *“Today you measured chance with fractions, found the hidden unfairness in a game, and let the computer play a thousand games to test your maths. Close, not exact: that is how random things behave.”*

## Looking back (2 minutes, minutes 86 to 88)

**SAY** *"Two minutes, just for you, in the "Looking back" box on the Finish page. Finish: something I did well, the hardest part was, and tick what helped you. You can write "pass". If you are still stuck, tick that box and I will check in with you in the open session, in the shared room, with your parent nearby. Nobody marks this and nobody has to show it."*

**DO** Everyone reaches this box: it is on the Finish page, not on the Think harder pages. Collect nothing in front of the group and never read an answer out. You may look at a child's box on camera in the open session if the child offers it. Praise the strategy, not the speed: "you you listed every outcome, and that is what got you unstuck", never "you were quick".

If a child ticks "I am still stuck", check in during the open session the same day: visit the shared breakout room or use a hand signal, with a parent nearby. Never by private message. If a child writes "everything" or "I am bad at maths", say quietly in the open session: "Thank you for telling me. Let's find one small thing that worked today," and point to a specific item they got right. If a child writes "nothing was hard", ask what would make it harder.

**DO** Minutes 88 to 90: stamp the inquiry journal when a child shows you the exit slip. Mark S, D or H in the progress tracker at the back of this guide. If a child marked D or H, plan two minutes of catch-up at the start of the next lesson.

## Extra idea for anyone who wants it

Change the Dice Race so that it is fair, using totals only (a total of 7 may mean "roll again"). Test your rules in code with 1000 games. How close to 500 : 500 do you get?

## Answer key (workbook)

**Q1** (a) 3/8 (b) 6/8 = 3/4 (c) about 375 (3/8 × 1000) (d) No. Each spin is random, so the count wobbles around 375; 362 is close. About 19 runs in 20 land between 345 and 405.

**Q2** (a) about 500 (1/2 × 1000) (b) No. The flips are random, so each run gives a slightly different count near 500. Exactly 500 happens only about 1 run in 40. (c) about 200 (1/5 × 1000)

**Q3** (a) 4/10 = 2/5 (b) 2 red (6 red, 6 blue: 6/12 = 1/2) (the "add blue to make 1/5" part is now in Think harder 4)

**Q4** (a) Differences: 0 appears 6 times, 1: 10, 2: 8, 3: 6, 4: 4, 5: 2 (36 in all). (b) Ali 24/36 = 2/3, Ben 12/36 = 1/3 (c) Not fair: Ali is twice as likely to win, even though each has 3 differences. (Making it fair is now in Think harder 4.)

**Q5** Gaps: 6, 6, a = b. (a) About 167: P(double) = 6/36 = 1/6, and 1/6 × 1000 = 166 2/3. (b) A count must be a whole number, and each run is random, so it lands near 167 (about 19 runs in 20 between 145 and 190), not exactly on it.

**Investigation** Ali is wrong. (1) Coin: 1/2 and 1/2, fair. (2) Dice: six has 1/6, not six has 5/6, unfair. (3) Two dice: a total of 7 has 6 out of 36 = 1/6, not 7 has 30 out of 36 = 5/6, unfair. Two outcomes is not the same as two EQUAL chances. Fair means every player has the same probability. Fixes: dice, 1 to 3 against 4 to 6; two dice, even total against odd total (18 out of 36 each, checked by listing all 36). Look for: a list or table of outcomes for game 3, probabilities as fractions, and the idea that one counter-example (game 2) is enough.

# Lesson 7: Turtle Geometry

Week 4 (first lesson) • Angles in Polygons • Workbook: Lesson 7

| | |
|---|---|
| **Objective** | Use the idea that a turtle going once round a closed shape turns 360° altogether to find the turn angle (360° ÷ sides) and inside angle (180° – turn) of regular polygons, work backwards from an angle to the number of sides, extend to star polygons (turn = 360° × times round ÷ points), and test every prediction with a pen program in Scratch (or Python turtle). |
| **Thinking skills** | Turn (exterior) and inside (interior) angles, angles on a straight line, angle sum of a polygon, star polygons, predicting and tracing code, repeat loops and variables |
| **Materials** | A laptop or desktop with Scratch 3 open (scratch.mit.edu) and the Pen extension added, mini whiteboards, a protractor, plain or squared paper. |
| **Online set-up** | The tutor builds a "Turtle start" Scratch file beforehand (Pen extension added and the 4 set-up blocks already in place; no .sb3 file exists in this repository) and shares it through the approved channel, so children only open it. Pairs work in breakout rooms: one child drives Scratch and shares the screen, the other predicts, then they swap. Children hold drawings up to the camera. A tablet works for Scratch but is slower. Every drawing starts with the same 4 set-up blocks (erase all, go to x: -50 y: 100, point in direction 90, pen down) so shapes stay on the stage; turn blocks in Scratch show an arrow (turn ↻ is clockwise, our "turn right"). Block colours on our slides are simplified; in Scratch, find blocks by the category name on the left (Motion, Pen, Control), not by colour. |
| **Prepare** | Solve the worksheet yourself first. Set out one materials box per pair. Open the slides for this lesson and check the answers in the speaker notes. |

### 0–2 min Opening

**SAY** *"Today you are turtle pilots. A turtle with a pen can draw any shape, but only if you tell it exactly how far to turn at every corner."*

*"Our rule today: work the turn out with maths first, then test it in code. Notice it. Test it. Explain why."*

**DO** Greet each child by name. Show the lesson title slide. Read "Today we will…" aloud.

### 2–5 min Sneak-peek puzzles

**SAY** *"Two sneak-peek puzzles. They are ideas we have not learnt yet, so guessing is welcome and "I do not know" is welcome. Nobody is marked. Write your answers on your own paper. When I say "show", everyone holds their paper to the camera at the same moment."*

| Sneak-peek puzzle | Answer and what to listen for |
|---|---|
| | |

| | |
|---|---|
| Stand up. Face the door. Turn a quarter turn to your right (clockwise). Do that 4 times in a row. Which way are you facing at the end? What did your body do in total? | Facing the door again. Four quarter turns make one full turn (4 × 90° = 360°). Children who cannot stand can use an arm or a drawing of an arrow. |
| A turtle goes once round a closed shape and turns 40° at every corner (all the turns equal). How many corners does the shape have? | 9 corners. The turtle turns 360° altogether and each turn is 40°, so 360 ÷ 40 = 9. A guess is welcome. |

**DO** Puzzle 1 is an access puzzle: any child can try it with objects, a drawing or counting, so everyone gets a first win. Puzzle 2 is a thinking puzzle on the idea of today's lesson.

Papers are shown to the camera by everyone at the same moment, or to you only on camera. Never use private chat and never read wrong answers aloud. Reveal the answers to everyone at minute 5 and say: "Both of these are what we learn today."

Do not record the sneak-peek as a mark. Never announce who got which right, never say who will start on which page, and never use the word "advanced" or "top group".

Invite the WHOLE group, never one child: "When you have finished Start here, you can choose Think harder, Go further or the Investigation." Nobody is singled out. The point printed on page 1 ("After question N you may choose…") is the same for all. Any child can choose it, and any child can go back. Do not split a pair unless both children agree.

### 5–10 min Warm-up

**DO** Turtle says. Stand (or sit and use your arm as the turtle's nose) and face the screen (north). The tutor calls turns; you turn your body and say which way you face. Last one: one full turn in 3 equal turns. How big is each? (5 to 8 minutes. Keep it friendly: no racing or timers, and no points for being first.)

**SAY** *"Stand up, or sit and use your arm as the turtle's nose, and face your screen. Call that north. I am the programmer and you are the turtle. Turn right 90. Which way are you facing now?"*

*"Now: how can you make one full turn using 3 equal turns? How big is each turn?"*

**DO** At about minute 8, ask the quick questions on the warm-up slides. Use thumbs or a whisper to partners before taking answers. Stop at minute 10.

| Ask | Expect |
|---|---|
| Angles on a straight line add up to …? | 180° |
| How many degrees in a full turn? | 360° |
| Two angles of a triangle are 50° and 75°. What is the third? | 55° |
| What is 360 ÷ 8? | 45 |

| | |
|---|---|
| Face north. Turn right 90°, then right 180°. Which way do you face? | West |

### 10–25 min Explore (in pairs, with objects)

**SAY** *"Here is the square program: repeat 4, move 100, turn right 90. I want an equilateral triangle. What turn should I type? Write your prediction on your whiteboard before anyone runs it."*

**DO** 1. Add the Pen extension. Build the 4 set-up blocks, then repeat 4 [move 100, turn ↻ 90].

2. Change it to draw an equilateral triangle. Predict the turn first, then run it.

3. Now a regular pentagon (5 sides) and a regular hexagon (6 sides). Predict, then test.

4. Record sides, turn and "did it close?". What do sides × turn always make?

**SAY** *"Talk to your partner while you work. I will come round and ask you questions."*

**IF STUCK** Stuck: walk the shape yourself. Walk one side, then turn at the corner. How far did your body turn?

Typed turn 60 for the triangle? Run it and look. Does the turtle come home? How far has it turned altogether?

Finished early: predict the turn for a regular shape with 10 sides, then test it.

**DO** Go further (from minute 12, on the explore slide, for any pair that wants it): What if you want a regular shape with 8 sides (an octagon)? Predict the turn. Then test it in Scratch with repeat 8. What about 10 sides? Answer: 45° for 8 sides (360 ÷ 8) and 36° for 10 sides (360 ÷ 10). Nobody has to wait: a pair that has finished can pick this up.

**DO** At about minute 22, say "Pencils down" and bring the group back together. Take the three questions below, one at a time.

| Ask | Expect |
|---|---|
| What turn did the triangle need? What happened if you typed 60? | 120°. Turn 60 draws only half a hexagon: 3 × 60 = 180°, so the turtle does not come home. |
| What turns did the pentagon and hexagon need? | Pentagon 72°, hexagon 60°. |
| What do sides × turn always make? Why? | 360°: the turtle comes home facing the same way, so it has made one full turn. |

### 25–35 min Mini-lesson (you teach, they join in)

**DO** Show the mini-lesson slides. Think aloud as you work. Say what you notice, what you test and how you check. Children copy how you think.

**SAY** *"An octagon has 8 equal turns. To come home facing the same way the turtle turns 360° altogether, so each turn is 360 ÷ 8 = 45°. The turn and the inside angle sit on one straight line, so the inside angle is 180 − 45 = 135°."*

*"Working backwards: the inside angle is 174°, so the turn is 180 − 174 = 6°. The turns make 360°, so there are 360 ÷ 6 = 60 sides."*

*"For the star the turtle goes round twice: 720° altogether. 5 equal turns: 720 ÷ 5 = 144°. The angle at each point is 180 − 144 = 36°."*

| Worked example on the slides | Answer |
|---|---|
| Regular octagon: find the turn. Going round once, the turtle turns 360° altogether. 8 equal turns: 360 ÷ 8 = ? Inside angle = 180° − turn. Test it: repeat 8 [move 80, turn ↻ ?] | Turn 45°, inside angle 135°. |
| A 5-point star. repeat 5 [move 150, turn ↻ 144] 5 × 144° = 720°: the turtle goes round twice. Turn = 360° × times round ÷ points = 720 ÷ 5. Angle at each point = 180° − turn = ? | Turn 144°, 36° at each point. |

**SAY** *"The key idea today is: the 360° turn. Going once round a closed shape, the turtle turns 360° altogether. A regular polygon with n sides turns 360° ÷ n at each corner, and its inside angle is 180° − turn. Say it with me."*

**DO** Show the key idea slide at minutes 33 to 35. The "common mistakes" on that slide are for you, not the children. Watch for them during the worksheet:

- Using the inside angle as the turn: turn 60° for a triangle draws half a hexagon, and turn 108° for a pentagon does not close.
- Dividing 180 by the number of sides instead of 360, or forgetting that turn + inside angle = 180°.
- Stars: thinking the turns always add up to 360°. A star goes round more than once, so its turns add up to 720°, 1080° and so on.

**35–40 min Brain break**

**DO** Turtle dance. Walk a square on the spot: step, quarter turn, four times. Then a triangle: step, turn a third of a turn, three times. Then a hexagon.

Water. Toilet break for anyone who needs it. Keep it to five minutes.

**40–75 min Worksheet (pairs, then alone, then your choice)**

**DO** Minute 40: give out the workbook. Read the title of Lesson 7 aloud. Say "Pairs first. Talk before you write."

Follow the minute labels on the slides: short teaching slides come before page 1 and before page 2, pairs work together on page 1 (about 40 to 55), each child works alone on page 2 (about 55 to 72), and minutes 72 to 75 are the check together. Any child who has finished can choose Think harder, Go further or the Investigation (all printed in the workbook). The Finish page is for everyone.

**SAY** *"Pairs first. Read each question aloud and talk about it before you write. I am not looking for speed. I am looking for good thinking."*

**DO** Circulate and ask:

- Ask: “What did you predict before you ran it? Were you right?”
- Ask: “How far has your turtle turned altogether? How do you know?”
- Ask: “Which angle does the turtle turn: the inside angle or the outside one? Show me on the picture.”

**IF STUCK** Hint 1: “To come home facing the same way, the turtle makes one full turn: 360°.”

Hint 2: “The corners are equal, so share 360° equally between them.”

Hint 3: “The turn and the inside angle sit on one straight line: they add up to 180°.”

**DO** Think harder, Go further and the Investigation are open to every child, with no label. The Investigation is printed on the Finish page that every child gets. After question 3 (see the line printed on page 1) any child may choose to carry on or try Think harder. Never announce who is on which page. Do not split a pair unless both children agree.

## Stretch and challenge questions

| Question | Answer |
|---|---|
| Workbook Think harder 1: Each inside angle of a regular polygon is 4 times its turn angle. How many sides does the polygon have? | 10 (turn + 4 × turn = 180°, so the turn is 36° and 360 ÷ 36 = 10; check: inside 144° = 4 × 36°) |
| Workbook Think harder 2: Each inside angle of a regular polygon is 140°. What do all its inside angles add up to? | 1260° (turn 40°, so 9 sides; 9 × 140° = 1260°; check (9 – 2) × 180° = 1260°) |
| Workbook Think harder 3: Mei’s turtle draws a shape with 4 sides. Its first three turns are 70°, 100° and 90°. It comes home facing the same way after going round once. What is its last turn? What is the inside angle at that corner? | Last turn 100° (360 – 70 – 100 – 90); inside angle 80° (180 – 100) |
| Workbook Think harder 4: Plan a 20-point star that goes round 3 times (move 100). Find the total turning, the turn angle and the point angle. Which has sharper points, this star or the 9-point star in Q4? | Total 3 × 360° = 1080°, turn 1080 ÷ 20 = 54°, point angle 180° – 54° = 126°. The 9-point star (point angle 20°) has sharper points. |
| Workbook Go further 5: Design your own star. Choose the number of points p and how many times it goes round, k, so that the turn angle 360° × k ÷ p is a WHOLE number and p and k share no common factor. Give the turn angle, the point angle (180° – turn) and the code. Explain why your star closes after p repeats. | Open. Example: 8 points going round 3 times: turn = 360 × 3 ÷ 8 = 135°, point angle 45°; code repeat 8 [move 100, turn ↻ 135]. After 8 repeats it has turned 8 × 135° = 1080° = 3 full turns, so it faces the way it started and the star closes. If p and k share a factor, the star closes sooner and is a smaller shape repeated (9 points going round 3 times is just a triangle). |

| | |
|---|---|
| Workbook Go further 6: A square, a hexagon and a 12-sided shape fit exactly around a point: 90° + 120° + 150° = 360°. Find one other set of three regular polygons whose inside angles add up to 360° (hint: the answer uses a square or a triangle). | Square + octagon + octagon (90° + 135° + 135°) or triangle + 12-gon + 12-gon (60° + 150° + 150°). |
| Workbook Go further 7: Try turn 120 and turn 60 with repeat 6 and move 100. Which one draws a hexagon? Why does the other not make a 6-point star? | Turn 60 draws the hexagon (6 × 60° = 360°, one full turn). Turn 120 draws a triangle twice round (6 × 120° = 720°, two full turns), and it only visits 3 corners, not 6. |
| Workbook Investigation, Finish page (What if we change ...? What stays the same?): A turtle repeats [move 100, turn ↻ t] until it first faces the way it started. Make a table for t = 40, 72, 90, 100, 120, 144 and 150: how many repeats, and how many times does it go round? Add up the turns to check. What changes when t changes? What stays the same? First step: a turtle repeats [move 100, turn 90]. How many repeats until it faces the way it started? | Repeats and times round: 40: 9 and 1; 72: 5 and 1; 90: 4 and 1; 100: 18 and 5; 120: 3 and 1; 144: 5 and 2; 150: 12 and 5 (every row checked by brute force). Changes: the number of repeats and the shape (polygon or star). Stays the same: the turns always add up to a whole number of full turns, repeats × t = 360 × times round. When t divides 360 the turtle goes round once and draws a polygon; otherwise it goes round more than once and draws a star. Look for: adding the turns until the total is a multiple of 360, a table, and the link repeats = 360 × rounds ÷ t. Children do not need to test every row in Scratch: a few are enough. |

These are the questions on the Think harder, Go further and Finish pages in the workbook. More are in the separate Question Bank.

### 75–82 min Talk time

**DO** Bring the group back to a circle (or camera on, mics on). Ask for volunteers first and choose two or three of them, not only the fastest. Never cold-call a child by name. “Pass” is always fine, and the next person helps. No public tallies, points or rankings.

**SAY** *“Ali says the turtle turns by the inside angle of the shape. Mei says it turns by the outside angle. Who is right? Use a triangle to explain.”*

*“Why do the turns add up to 360° for every shape the turtle draws going round once?”*

| Discussion question | Listen for |
|---|---|
| Ali says the turtle turns by the inside angle of the shape. Mei says it turns by the outside angle. Who is right? Use a triangle to explain. | Mei is right. For an equilateral triangle the inside angle is 60° but the turtle turns 120°; turning 60° three times only draws half a hexagon. The turtle starts facing one way and comes home facing the same way after going round once, so it has made exactly one full turn: 360°. A star goes round more than once, so its turns add up to 720°, 1080° and so on. |

### 82–90 min Exit slip, looking back and closing

**DO** Minutes 82 to 86: exit slip. Children write on the exit slip box of the Finish page (or a slip of paper). No talking, no helping. Show the exit ticket slide.

| Exit slip | Answer |
|---|---|
| A turtle draws a regular polygon and turns 5° at every corner. How many sides does the polygon have? What is each inside angle? | 72 sides (360 ÷ 5 = 72); each inside angle is 175° (180 − 5) |

**SAY** *"Today you predicted every turn with maths and then tested it in code. That is how mathematicians and programmers work together. Well done, inquirers!"*

## Looking back (2 minutes, minutes 86 to 88)

**SAY** *"Two minutes, just for you, in the "Looking back" box on the Finish page. Finish: something I did well, the hardest part was, and tick what helped you. You can write "pass". If you are still stuck, tick that box and I will check in with you in the open session, in the shared room, with your parent nearby. Nobody marks this and nobody has to show it."*

**DO** Everyone reaches this box: it is on the Finish page, not on the Think harder pages. Collect nothing in front of the group and never read an answer out. You may look at a child's box on camera in the open session if the child offers it. Praise the strategy, not the speed: "you you added up the turns, and that is what got you unstuck", never "you were quick".

If a child ticks "I am still stuck", check in during the open session the same day: visit the shared breakout room or use a hand signal, with a parent nearby. Never by private message. If a child writes "everything" or "I am bad at maths", say quietly in the open session: "Thank you for telling me. Let's find one small thing that worked today," and point to a specific item they got right. If a child writes "nothing was hard", ask what would make it harder.

**DO** Minutes 88 to 90: stamp the inquiry journal when a child shows you the exit slip. Mark S, D or H in the progress tracker at the back of this guide. If a child marked D or H, plan two minutes of catch-up at the start of the next lesson.

## Extra idea for anyone who wants it

Draw a rosette: repeat 12 [draw a square, then turn ↻ 30]. Start this one with go to x: 0 y: 0 instead of x: -50 y: 100, so the rosette stays on the stage. How far does the turtle turn altogether? (12 × (360 + 30) = 4680° = 13 full turns.) Design your own rosette and predict its total turning before you run it.

## Answer key (workbook)

**Q1** 3 sides (example): 120°, 60°, 360°. 9 sides: turn 40°, inside 140°. 10 sides: 36°, 144°. 36 sides: 10°, 170°. Turn 24°: 15 sides, inside 156°. Inside 165°: turn 15°, 24 sides. Last column 360° every time. (a) Sides × turn = 360°: the turtle makes one full turn to come home facing the same way. (b) No: the turn would be 50°, and 360 ÷ 50 = 7.2 is not a whole number of sides.

**Q2** (a) An equilateral triangle, drawn 3 times over (each turn of 120° is a triangle's turn, so after 3 moves the turtle is back where it started). (b) 9 × 120° = 1080° = 3 full turns. (c) Change the turn from 120 to 40 (360 ÷ 9).

**Q3** Totals: 80, 160, 240, 320, 400, 480, 560, 640, 720. (a) 720 (b) 2 times (720 ÷ 360) (c) a 9-point star (d) 100° (180 – 80)

**Q4** 9 points, 4 times round: total 1440°, turn 160°, point 20°. (The 20-point star, total 1080°, turn 54°, point 126°, is now Think harder 4.)

**Q5** Each inside angle: 60°, 90°, 108°, 120°, 144°. Added: 180°, 360°, 540°, 720°, 1440°. (a) 180° (b) (n – 2) × 180°, or n × 180° – 360°

**Investigation** Repeats and times round: 40: 9 and 1; 72: 5 and 1; 90: 4 and 1; 100: 18 and 5; 120: 3 and 1; 144: 5 and 2; 150: 12 and 5 (every row checked by brute force). Changes: the number of repeats and the shape (polygon or star). Stays the same: the turns always add up to a whole number of full turns, repeats × t = 360 × times round. When t divides 360 the turtle goes round once and draws a polygon; otherwise it goes round more than once and draws a star. Look for: adding the turns until the total is a multiple of 360, a table, and the link repeats = 360 × rounds ÷ t. Children do not need to test every row in Scratch: a few are enough.

# Lesson 8: Secret Codes

Week 4 (second lesson) • Clock (Modular) Arithmetic • Workbook: Lesson 8

| | |
|---|---|
| **Objective** | Do clock arithmetic (counting past the end wraps round to the start), use n mod m as the remainder when n is divided by m, solve clock and calendar puzzles with mod 12 and mod 7, code and decode Caesar ciphers with letter numbers (A = 0 … Z = 25, code = (letter + shift) mod 26), crack a code, and test it all with a Scratch program that uses a list of letters and mod. |
| **Thinking skills** | Remainders and mod, clock and calendar arithmetic, negative numbers on a clock, Caesar ciphers, cracking codes by reasoning, lists and letters in code, tracing code in a table |
| **Materials** | A laptop or desktop with Scratch 3 open (the tutor builds and shares a starter project with a list called alphabet holding A to Z; no .sb3 file exists in this repository, so build it beforehand), mini whiteboards, paper for code strips, scissors. |
| **Online set-up** | Children make their code strips on paper and hold them to the camera. Children hold up their coded name on paper to the camera, or the tutor reads out the coded names; partners decode on paper. Pairs use breakout rooms for the coding part; the driver shares the screen. Starter project: the tutor shares an .sb3 file (not a link only) containing the list alphabet with A to Z already filled in, and the loop already built; children add the three cipher blocks. Block colours on our slides are simplified; in Scratch, list blocks are red-orange (Variables palette) and you find blocks by the category name on the left, not by colour. |
| **Prepare** | Solve the worksheet yourself first. Set out one materials box per pair. Open the slides for this lesson and check the answers in the speaker notes. |

### 0–2 min Opening

**SAY** *"Today you are code-breakers. Spies, generals and computers all hide messages with codes. The oldest one we know is over 2000 years old, and it runs on the same maths as a clock."*

*"Notice it. Test it. Explain why. Today "test it" means: crack it, then let the computer check."*

**DO** Greet each child by name. Show the lesson title slide. Read "Today we will…" aloud.

### 2–5 min Sneak-peek puzzles

**SAY** *"Two sneak-peek puzzles. They are ideas we have not learnt yet, so guessing is welcome and "I do not know" is welcome. Nobody is marked. Write your answers on your own paper. When I say "show", everyone holds their paper to the camera at the same moment."*

| Sneak-peek puzzle | Answer and what to listen for |
|---|---|
| A clock shows 8 o'clock. What time is it 5 hours later? Use a clock drawing or count on with your fingers. | 1 o'clock (8, 9, 10, 11, 12, 1). Counting on round the clock face works. |

| | |
|---|---|
| Today is Wednesday. What day of the week will it be in 100 days? What do you think? How could you find out? | Friday. 100 ÷ 7 leaves remainder 2 (98 is 14 full weeks), so move 2 days on from Wednesday: Thursday, Friday. Guessing and “I do not know” are fine. |

**DO** Puzzle 1 is an access puzzle: any child can try it with objects, a drawing or counting, so everyone gets a first win. Puzzle 2 is a thinking puzzle on the idea of today’s lesson.

Papers are shown to the camera by everyone at the same moment, or to you only on camera. Never use private chat and never read wrong answers aloud. Reveal the answers to everyone at minute 5 and say: “Both of these are what we learn today.”

Do not record the sneak-peek as a mark. Never announce who got which right, never say who will start on which page, and never use the word “advanced” or “top group”.

Invite the WHOLE group, never one child: “When you have finished Start here, you can choose Think harder, Go further or the Investigation.” Nobody is singled out. The point printed on page 1 (“After question N you may choose…”) is the same for all. Any child can choose it, and any child can go back. Do not split a pair unless both children agree.

### 5–10 min Warm-up

**DO** Clock jumps. The tutor shows a clock and calls a time and a jump (“9 o’clock, 4 hours later?”). Answer on whiteboards. Then jumps past 12, jumps backwards, and a day-of-the-week jump. (5 to 8 minutes. Keep it friendly: no racing or timers, and no points for being first.)

**SAY** *“Look at the clock. It is 9 o’clock. What time will it be 4 hours later? How did you work it out?”*

*“Now 11 o’clock and 13 hours later. Did you go past 12? What happened?”*

**DO** At about minute 8, ask the quick questions on the warm-up slides. Use thumbs or a whisper to partners before taking answers. Stop at minute 10.

| Ask | Expect |
|---|---|
| 17 ÷ 5: what is the remainder? | 2 |
| 30 ÷ 7: what is the remainder? | 2 |
| 26 ÷ 26: what is the remainder? | 0 |
| 40 ÷ 12: what is the remainder? | 4 |
| Which of 7, 9, 10, 12 leave remainder 1 when divided by 3? | 7 and 10 |

### 10–25 min Explore (in pairs, with objects)

**SAY** *“With a shift of 3, A becomes D and B becomes E. What do you think Z becomes? Write your guess before you make the strip.”*

**DO** 1. Make a code strip: write A to Z. Underneath, write the alphabet again starting at D (A → D, B → E …).

2. Code your first name with shift 3. Swap with your partner and decode each other's.

3. Decode the tutor's message: PDWK LV IXQ

4. What happens to X, Y and Z? How do you decode: which way do you shift?

**SAY** *"Talk to your partner while you work. I will come round and ask you questions."*

**IF STUCK** Stuck making the strip: write A to Z on one line. Underneath, start at D under A, and keep going.

Stuck at the end of the alphabet: "After Z, where does the bottom row carry on?"

Finished early: invent a shift, code a short message for your partner, and give them only one clue.

**DO** Go further (from minute 12, on the explore slide, for any pair that wants it): What if the shift is 13? Code the word HELLO with shift 13. Then code your answer with shift 13 again. What do you notice? Predict first. Answer: HELLO becomes URYYB, and coding URYYB with shift 13 gives HELLO again. Two shifts of 13 make 26, a full circle of the alphabet. Nobody has to wait: a pair that has finished can pick this up.

**DO** At about minute 22, say "Pencils down" and bring the group back together. Take the three questions below, one at a time.

| Ask | Expect |
|---|---|
| What do X, Y and Z become with shift 3? | A, B and C: after Z the alphabet wraps round to the start. |
| How do you decode a shift-3 message? | Shift each letter back 3 (or forward 23). PDWK LV IXQ is MATH IS FUN. |
| What does a shift of 26 do? | Nothing: every letter goes once round and lands on itself. |

**25–35 min Mini-lesson (you teach, they join in)**

**DO** Show the mini-lesson slides. Think aloud as you work. Say what you notice, what you test and how you check. Children copy how you think.

**SAY** *"It is 9 o'clock. I count on 5 hours: 10, 11, 12, 1, 2. Adding gives 14, but a clock wraps round after 12, so 14 − 12 = 2. In maths we write 14 mod 12 = 2: the remainder when 14 is divided by 12."*

*"To code Z with shift 5: Z is 25. 25 + 5 = 30. There is no letter 30, so I wrap round: 30 − 26 = 4, and 4 is E."*

*"To decode A with shift 4: A is 0. 0 − 4 = −4. Below zero, so I add 26: −4 + 26 = 22, and 22 is W."*

| Worked example on the slides | Answer |
|---|---|
| Clock arithmetic: 9 o'clock + 5 hours. Count on 5: 10, 11, 12, 1, 2. Adding gives 14. A clock wraps round after 12: 14 − 12 = 2. 14 mod 12 = 2: the remainder when 14 is divided by 12. Thursday + 45 days: 45 mod 7 = ? | 2 o'clock. 45 mod 7 = 3, so Sunday. |

| | |
|---|---|
| Caesar code: ZOO with shift 5. Letter numbers: Z = 25, O = 14. Z: 25 + 5 = 30, 30 mod 26 = 4 = E. O: 14 + 5 = 19 = T. Decode AEB (shift 4): A = 0, 0 − 4 = −4, + 26 = 22 = W … | ZOO → ETT. AEB → WAX. |

**SAY** *"The key idea today is: wrap round like a clock. n mod m is the remainder when n is divided by m. On a clock with m numbers, counting past the end wraps round to the start. Say it with me."*

**DO** Show the key idea slide at minutes 33 to 35. The "common mistakes" on that slide are for you, not the children. Watch for them during the worksheet:

- Off by one: counting A as 1 in the maths (we use A = 0, Z = 25), or forgetting that Scratch's list starts at item 1.
- Not wrapping round: writing a number bigger than 25 (or below 0 when decoding) instead of taking away or adding 26.
- Calendars: counting today as day 1 (Saturday + 2 days is Monday, not Sunday), or giving the quotient instead of the remainder for mod.

**35–40 min Brain break**

**DO** Clock arms. Your arm is the hour hand. The tutor calls a time and you point to it. Then "add 5 hours!" and you sweep your arm round to the new time. Try "add 12 hours!"

Water. Toilet break for anyone who needs it. Keep it to five minutes.

**40–75 min Worksheet (pairs, then alone, then your choice)**

**DO** Minute 40: give out the workbook. Read the title of Lesson 8 aloud. Say "Pairs first. Talk before you write."

Follow the minute labels on the slides: short teaching slides come before page 1 and before page 2, pairs work together on page 1 (about 40 to 55), each child works alone on page 2 (about 55 to 72), and minutes 72 to 75 are the check together. Any child who has finished can choose Think harder, Go further or the Investigation (all printed in the workbook). The Finish page is for everyone.

**SAY** *"Pairs first. Read each question aloud and talk about it before you write. I am not looking for speed. I am looking for good thinking."*

**DO** Circulate and ask:

- Ask: "What number is that letter? Did you start at A = 0?"
- Ask: "Your number went past 25. What do you do?"
- Ask: "How can you check your code? Decode it again and see if you get back the message."

**IF STUCK** Hint 1: "Change each letter to its number first. Use the strip."

Hint 2: "Add the shift. If you go past 25, take away 26."

Hint 3: "To decode, take away the shift. If you go below 0, add 26."

**DO** Think harder, Go further and the Investigation are open to every child, with no label. The Investigation is printed on the Finish page that every child gets. After question 3 (see the line

printed on page 1) any child may choose to carry on or try Think harder. Never announce who is on which page. Do not split a pair unless both children agree.

## Stretch and challenge questions

| Question | Answer |
|---|---|
| Workbook Think harder 1: The pattern CODECODECODE... goes on and on. What is the 100th letter? | E (100 mod 4 = 0, so the 100th letter is the 4th letter of CODE) |
| Workbook Think harder 2: It is 10 o'clock now. What time will it be 1000 hours from now (on a 12-hour clock)? | 2 o'clock (1000 mod 12 = 4, and 10 + 4 = 14, 14 − 12 = 2) |
| Workbook Think harder 3: Find the smallest number bigger than 10 that leaves a remainder of 3 when it is divided by 4 and also when it is divided by 5. | 23 (3 more than a multiple of 20: 23 = 5 × 4 + 3 = 4 × 5 + 3) |
| Workbook Think harder 4: (a) Work out 29 mod 12, 50 mod 7 and 100 mod 26. (b) Decode the whole message from Q4 (BPM SMG QA CVLMZ BPM UIB) with the shift you found. | (a) 29 mod 12 = 5, 50 mod 7 = 1, 100 mod 26 = 22. (b) THE KEY IS UNDER THE MAT. |
| Workbook Go further 5: Write your own secret message of at least 8 letters and code it with a shift of your choice. Write clear instructions telling a friend how to decode it, and explain how a friend could crack it without being told the shift. Swap and decode. | Open. Check the code by decoding it, and that the cracking idea is sensible: try every shift from 1 to 25, or use the commonest letter (often E) or a short word (A or I). Example: MEET AT SIX with a shift of 3 is PHHW DW VLA. |
| Workbook Go further 6: Ali codes a message with a shift of 7. Mei codes Ali's result with another shift. The final message is the same as the original coded with a shift of 2. What shift did Mei use? | 21 (7 + 21 = 28, and 28 mod 26 = 2) |
| Workbook Go further 7: Start at 12 on a clock and keep jumping forward 5 hours. Do you land on every number before you get back to 12? What about jumps of 4 hours? | Jumps of 5: yes, all 12 numbers (5, 10, 3, 8, 1, 6, 11, 4, 9, 2, 7, 12). Jumps of 4: no, only 4, 8 and 12, because 4 shares a factor with 12. |

| | |
|---|---|
| Workbook Investigation, Finish page (Is it always true? Test and explain): Code the word CAT with a shift of 8. Then code your answer with a shift of 18. What do you get? Try two other shifts that add up to 26. (a) Is it always true that two such shifts undo each other? Explain. (b) Is there a shift that is its own decoder, so that coding twice gives back the original? Find ALL of them. First step: code the letter C with shift 8. Then shift your answer by 18. Where do you land? | CAT with shift 8 is KIB (C 2 → 10, A 0 → 8, T 19 → 27 mod 26 = 1). KIB with shift 18 is CAT again. (a) Yes: shifting by a and then by 26 − a moves every letter 26 places in total, a full circle round the alphabet, so each letter lands where it started. (b) Only shift 13: 13 + 13 = 26. No other shift from 1 to 25 works because 2 × shift would have to be a multiple of 26 (checked by brute force). Look for: the two worked examples, the “26 is a full circle” idea, and a systematic check for (b). Note 13 is the code known as ROT13. |

These are the questions on the Think harder, Go further and Finish pages in the workbook. More are in the separate Question Bank.

### 75–82 min Talk time

**DO** Bring the group back to a circle (or camera on, mics on). Ask for volunteers first and choose two or three of them, not only the fastest. Never cold-call a child by name. “Pass” is always fine, and the next person helps. No public tallies, points or rankings.

**SAY** *“Why does a shift of 26 change nothing?”*

*“Ali says decoding shift 3 is the same as coding again with shift 23. Is he right? Why?”*

| Discussion question | Listen for |
|---|---|
| Why does a shift of 26 change nothing? Why is decoding a shift-3 message the same as coding it again with shift 23? | A shift of 26 moves every letter once round the whole 26-letter clock, so each letter lands back on itself (26 mod 26 = 0). Ali is right: shifting back 3 and shifting forward 23 land on the same letter, because 23 + 3 = 26 is a full turn. For example D (3): 3 − 3 = 0 = A, and 3 + 23 = 26, 26 mod 26 = 0 = A. |

### 82–90 min Exit slip, looking back and closing

**DO** Minutes 82 to 86: exit slip. Children write on the exit slip box of the Finish page (or a slip of paper). No talking, no helping. Show the exit ticket slide.

| Exit slip | Answer |
|---|---|
| Code the word HI with a shift of 25. What do you notice? | GH (H = 7, 7 + 25 = 32, 32 mod 26 = 6 = G; I = 8, 33 mod 26 = 7 = H). A shift of 25 is the same as shifting back 1, because 25 = 26 − 1. |

**SAY** *“Today you used clock arithmetic to make and break codes, and you made the computer do it for you. Mathematicians call this modular arithmetic, and it helps protect messages sent on the internet. Well done, code-breakers!”*

## Looking back (2 minutes, minutes 86 to 88)

**SAY** *"Two minutes, just for you, in the "Looking back" box on the Finish page. Finish: something I did well, the hardest part was, and tick what helped you. You can write "pass". If you are still stuck, tick that box and I will check in with you in the open session, in the shared room, with your parent nearby. Nobody marks this and nobody has to show it."*

**DO** Everyone reaches this box: it is on the Finish page, not on the Think harder pages. Collect nothing in front of the group and never read an answer out. You may look at a child's box on camera in the open session if the child offers it. Praise the strategy, not the speed: "you you wrote out the alphabet, and that is what got you unstuck", never "you were quick".

If a child ticks "I am still stuck", check in during the open session the same day: visit the shared breakout room or use a hand signal, with a parent nearby. Never by private message. If a child writes "everything" or "I am bad at maths", say quietly in the open session: "Thank you for telling me. Let's find one small thing that worked today," and point to a specific item they got right. If a child writes "nothing was hard", ask what would make it harder.

**DO** Minutes 88 to 90: stamp the inquiry journal when a child shows you the exit slip. Mark S, D or H in the progress tracker at the back of this guide. If a child marked D or H, plan two minutes of catch-up at the start of the next lesson.

## Extra idea for anyone who wants it

The starter already keeps spaces (if the letter is in the alphabet list, code it, else join the letter itself). Add a "crack" button that says the message decoded with every shift from 1 to 25.

## Answer key (workbook)

**Q1** (a) 3 o'clock (15 − 12) (b) 5 o'clock (30 mod 12 = 6, 11 + 6 = 17, 17 − 12 = 5) (c) 7 o'clock (4 − 9 = −5, −5 + 12 = 7) (d) Monday (100 mod 7 = 2, Saturday + 2) (e) Thursday (30 mod 7 = 2, Saturday back 2) (the three "mod" sums are now in Think harder 4)

**Q2** (a) WPGGH (b) CODE IS FUN (c) 16 (10 + 16 = 26)

**Q3** W: 22, 27, 1, B. A: 0, 5, 5, F. Y: 24, 29, 3, D. The program says BFD.

**Q4** (a) BPM, probably THE (b) 8 (T = 19, 19 + 8 = 27, 27 mod 26 = 1 = B) (c) THE (the first word). The whole message, THE KEY IS UNDER THE MAT, is now in Think harder 4.

**Q5** (a) 15 (b) 5 (31 mod 26) (c) He gets the original message back: 9 + 17 = 26, a full turn (d) 13 (13 + 13 = 26)

**Investigation** CAT with shift 8 is KIB (C 2 → 10, A 0 → 8, T 19 → 27 mod 26 = 1). KIB with shift 18 is CAT again. (a) Yes: shifting by a and then by 26 − a moves every letter 26 places in total, a full circle round the alphabet, so each letter lands where it started. (b) Only shift 13: 13 + 13 = 26. No other shift from 1 to 25 works because 2 × shift would have to be a multiple of 26 (checked by brute force). Look for: the two worked examples, the "26 is a full circle" idea, and a systematic check for (b). Note 13 is the code known as ROT13.

# Lesson 9: Data Detectives

Week 5 (first lesson) • Averages and Data • Workbook: Lesson 9

| | |
|---|---|
| **Objective** | Find the mean as total ÷ number of values and as levelling out, use total = mean × number of values to find a missing value, predict how changing the data changes the mean, compare the mean with the median and mode when one value is extreme, and build a list program that adds up the data and finds the mean. |
| **Thinking skills** | Mean, working back from a mean to a total, reasoning about change, median and mode, lists and loops in code, tracing code |
| **Materials** | Laptop or desktop with Scratch 3 (scratch.mit.edu; free, in the browser), workbook, pencil, mini whiteboard, about 25 linking cubes or counters for levelling towers. A calculator only for checking. |
| **Online set-up** | Data is collected in the chat. Children share their screen to show their code. The tutor shares a starter project as an .sb3 file (one sprite with an empty list called data; children use File, Load from your computer) or children build it from the slide in 5 minutes. Block colours on our slides are simplified; in Scratch, find blocks by the category name on the left, not by colour. The mean is familiar from primary school; the new parts are lists and loops, median and mode, and misleading averages. |
| **Prepare** | Solve the worksheet yourself first. Set out one materials box per pair. Open the slides for this lesson and check the answers in the speaker notes. |

### 0–2 min Opening

**SAY** *"Welcome back, data detectives! Newspapers, games and sports scores all come with averages. Today we find out what the mean really is, how one number can trick it, and how a computer adds up a whole list in a blink."*

*"Our rule today: work it out by hand first, then let the code test it. Notice it. Test it. Explain why."*

**DO** Greet each child by name. Show the lesson title slide. Read "Today we will…" aloud.

### 2–5 min Sneak-peek puzzles

**SAY** *"Two sneak-peek puzzles. They are ideas we have not learnt yet, so guessing is welcome and "I do not know" is welcome. Nobody is marked. Write your answers on your own paper. When I say "show", everyone holds their paper to the camera at the same moment."*

| Sneak-peek puzzle | Answer and what to listen for |
|---|---|
| Four friends have 2, 4, 4 and 6 stickers. Use counters to share the stickers out so that every friend has the same number. How many does each friend get? | 4 each. Total 16 stickers, shared between 4 friends. (This is the "fair share" idea behind the mean.) |

| | |
|---|---|
| Five friends have $2, $3, $3, $4 and $38. If they shared it all equally, each would get $10. Is $10 a good description of what a typical friend has? Why or why not? | No: four of the five friends have $4 or less. One very large amount pulls the equal share up. The middle value ($3) says more about a typical friend. Guesses are welcome. |

**DO** Puzzle 1 is an access puzzle: any child can try it with objects, a drawing or counting, so everyone gets a first win. Puzzle 2 is a thinking puzzle on the idea of today's lesson.

Papers are shown to the camera by everyone at the same moment, or to you only on camera. Never use private chat and never read wrong answers aloud. Reveal the answers to everyone at minute 5 and say: "Both of these are what we learn today."

Do not record the sneak-peek as a mark. Never announce who got which right, never say who will start on which page, and never use the word "advanced" or "top group".

Invite the WHOLE group, never one child: "When you have finished Start here, you can choose Think harder, Go further or the Investigation." Nobody is singled out. The point printed on page 1 ("After question N you may choose…") is the same for all. Any child can choose it, and any child can go back. Do not split a pair unless both children agree.

### 5–10 min Warm-up

**DO** Quick-eye mean. The tutor flashes five numbers for 5 seconds. Write your best estimate of the mean, no working out. Then we talk about how you guessed. No points. Three rounds. (5 to 8 minutes. Keep it friendly: no racing or timers, and no points for being first.)

**SAY** *"Quick-eye mean! I will show five numbers for five seconds only. No working out: just write your best guess of the mean. This is a guess, so any estimate is welcome. There are no points."*

*"How did you guess so fast? Who looked for a number in the middle and checked that the big ones and small ones balance?"*

**DO** At about minute 8, ask the quick questions on the warm-up slides. Use thumbs or a whisper to partners before taking answers. Stop at minute 10.

| Ask | Expect |
|---|---|
| Find the mean of 4, 6 and 8. | 6 |
| Find the mean of 10, 20, 30 and 40. | 25 |
| 5 children have a mean of 12 sweets. How many sweets altogether? | 60 |
| Find the mean of 3, 3, 3, 3 and 100. | 22.4 |
| Find the mean of 2.5 and 3.5. | 3 |

### 10–25 min Explore (in pairs, with objects)

**SAY** *"Before we add anything up: the mean number of letters in our names. Will it be bigger or smaller than the number of letters in YOUR name? Write a prediction."*

*"If our tutor joins with a 30-letter name, will the mean go up a little or a lot? Thumbs up for a lot, sideways for a little."*

**DO** 1. Everyone posts in the chat the number of letters in their full name (no spaces). Write the list.

2. Estimate the mean. Then work it out by hand: total ÷ how many.

3. Type the data into the Scratch list called data and run the mean program. Same answer?

4. Predict, then test: add a value of 30 to the list. What happens to the mean?

**SAY** *"Talk to your partner while you work. I will come round and ask you questions."*

**IF STUCK** Stuck on the hand calculation: add the values two at a time and write each running total, just like the variable total in the code.

Code says something strange: check that every value is in the list once, and that you divide by the length of the list, not by a number you typed.

Finished early: find a new value you could add that keeps the mean exactly the same. Test it in the program.

**DO** Go further (from minute 12, on the explore slide, for any pair that wants it): What if you add a very small value, 1, to the list instead of 30? Predict: does the mean go up or down? Does the median change? Test it and explain. Answer: The mean goes down (1 is below the other values, which are name lengths of about 8 or more). The median moves only to the next value along, so it changes by much less than the mean. One extreme value pulls the mean more than the median. Nobody has to wait: a pair that has finished can pick this up.

**DO** At about minute 22, say "Pencils down" and bring the group back together. Take the three questions below, one at a time.

| Ask | Expect |
|---|---|
| Was your estimate close? How did you estimate? | Pick a value in the middle, then check that the values above it and below it roughly balance. |
| What happened to the mean when 30 was added? | It went up, a lot, because 30 is far above every other value. One big value pulls the mean towards it. |
| Did the program and your hand calculation agree? | They should. If not, look for a value typed twice or missed, or dividing by the wrong number. |

**25–35 min Mini-lesson (you teach, they join in)**

**DO** Show the mini-lesson slides. Think aloud as you work. Say what you notice, what you test and how you check. Children copy how you think.

**SAY** *"Goals in 5 matches: 3, 0, 4, 1, 2. The 0 is a real value: that match counts. Total 3 + 0 + 4 + 1 + 2 = 10. Five matches, so the mean is 10 ÷ 5 = 2. Check by balancing: 3 is 1 above, 4 is 2 above: 3 above in all. 0 is 2 below, 1 is 1 below: 3 below in all. They balance, so 2 is right."*

*"The mean of 4 test scores is 15, so the total must be 4 × 15 = 60. The three I know make 12 + 18 + 14 = 44. The missing score is 60 − 44 = 16. I never subtract from the mean: I subtract from the*

*total."*

*"Goals by 7 players: 1, 2, 2, 3, 4, 5, 25. The mean is 42 ÷ 7 = 6, but 6 of the 7 players scored fewer than 6! The one big value pulled the mean up. The median, the middle value in order, is 3. That describes a typical player better."*

| Worked example on the slides | Answer |
|---|---|
| Find the mean, then balance-check. Goals in 5 matches: 3, 0, 4, 1, 2. Total: 3 + 0 + 4 + 1 + 2 = 10. There are 5 values (the 0 counts). Mean = 10 ÷ 5. Check: above the mean and below the mean should balance. | 2. Above: +1 and +2. Below: −2 and −1. They balance. |
| Missing value: use the total. The mean of 4 test scores is 15. Three of them are 12, 18 and 14. Total of all 4: 4 × 15 = 60. Known: 12 + 18 + 14 = 44. Missing score = 60 − 44. | 16. A 5th score of 15 would keep the mean at 15. |

**SAY** *"The key idea today is: mean = total ÷ how many. Mean = total ÷ how many values. Turn it round: total = mean × how many. The values above and below the mean always balance. Say it with me."*

**DO** Show the key idea slide at minutes 33 to 35. The "common mistakes" on that slide are for you, not the children. Watch for them during the worksheet:

- Dividing by the wrong count: leaving out a value of 0, or forgetting the new value when one is added.
- Finding a missing value from the MEAN instead of from the TOTAL (mean × how many) minus the known values.
- Thinking the mean must be one of the values, or that it is always a typical value even when one value is extreme.

**35–40 min Brain break**

**DO** Dice jumps. Roll two dice on screen: everyone does that many star jumps. Three rounds. What was the mean number of jumps per round?

Water. Toilet break for anyone who needs it. Keep it to five minutes.

**40–75 min Worksheet (pairs, then alone, then your choice)**

**DO** Minute 40: give out the workbook. Read the title of Lesson 9 aloud. Say "Pairs first. Talk before you write."

Follow the minute labels on the slides: short teaching slides come before page 1 and before page 2, pairs work together on page 1 (about 40 to 55), each child works alone on page 2 (about 55 to 72), and minutes 72 to 75 are the check together. Any child who has finished can choose Think harder, Go further or the Investigation (all printed in the workbook). The Finish page is for everyone.

**SAY** *"Pairs first. Read each question aloud and talk about it before you write. I am not looking for speed. I am looking for good thinking."*

**DO** Circulate and ask:

- Ask: "What is the total? How many values are there? Did you count the 0?"
- Ask: "Before you calculate: will the mean go up, go down or stay the same? Why?"
- Ask: "Point to the block that adds up. What is total after the second time round the loop?"

**IF STUCK** Hint 1: "Find the total first. Mean × how many gives the total."

Hint 2: "Draw the values as towers. Move cubes from tall towers to short ones until they are level."

Hint 3: "For a trace table, do one loop at a time: which item? what is total now? then change i by 1."

**DO** Think harder, Go further and the Investigation are open to every child, with no label. The Investigation is printed on the Finish page that every child gets. After question 3 (see the line printed on page 1) any child may choose to carry on or try Think harder. Never announce who is on which page. Do not split a pair unless both children agree.

## Stretch and challenge questions

| Question | Answer |
|---|---|
| Workbook Think harder 1: The mean height of 4 children is 150 cm. A fifth child joins the group and the mean height becomes 152 cm. How tall is the fifth child? | 160 cm (5 × 152 = 760, 4 × 150 = 600, 760 − 600 = 160) |
| Workbook Think harder 2: Mei's mean score for 3 tests is 84. What must she score in the 4th test to raise her mean to 86? | 92 (4 × 86 = 344, 3 × 84 = 252, 344 − 252 = 92) |
| Workbook Think harder 3: Write 5 whole numbers with a mean of 6 and a mode of 4. | Open, e.g. 4, 4, 5, 8, 9 (total 30, and 4 appears most often). Check the total is 30. |
| Workbook Think harder 4: A Scratch list holds 8 numbers with a mean of 12. Then the number 21 is added to the list. What will the mean program say now? | 13 (8 × 12 = 96, 96 + 21 = 117, 117 ÷ 9 = 13) |
| Workbook Go further 5: Make up your own "mean mystery" like the Think harder mean puzzles: choose five whole numbers, then write clues (any of mean, median, mode, smallest, largest) that give ONLY your numbers. Swap with a partner. Can they find a different set that fits your clues? | Open. Example clues: mean 5, median 4, mode 4, smallest 2, largest 9. The total is 25, so the other two numbers add to 25 − 2 − 4 − 9 = 10; only 4 and 6 fit (4 must appear twice to be the mode), giving 2, 4, 4, 6, 9. Check that the clues really give one set. |

| | |
|---|---|
| Workbook Go further 6: The mean of 1, 2, 3, … , 99 is 50. Explain why without adding them all up. Then plan a short program that checks it. | Pair them: 1 + 99, 2 + 98, … , 49 + 51 each make 100, with 50 left in the middle. Every pair balances around 50, so the mean is 50 (total 4950 ÷ 99 = 50). Program: set total to 0, set n to 1, repeat 99 (change total by n, change n by 1), say total ÷ 99. |
| Workbook Go further 7: A group of 8 children has a mean mass of 41 kg. When their tutor joins, the mean mass of all 9 becomes 44 kg. Then one more person joins and the mean of all 10 becomes 43 kg. What is the tutor's mass? What is the mass of the last person to join? | Tutor: 68 kg (9 × 44 = 396, 8 × 41 = 328, 396 − 328 = 68). Last person: 34 kg (10 × 43 = 430, 430 − 396 = 34). |
| Workbook Investigation, Finish page (What if we change …? What stays the same?): Data: 4, 6, 7, 9, 14. Find the mean, the median and the mode (is there one?). Now change the 14 to 24, then change it to 4, then to 7. Make a table of the mean, median and mode each time. What changes? What stays the same? Can you change TWO numbers and keep the mean the same? First step: find the mean of 4, 6, 7, 9 and 14 (add them up, then share equally). | Original: mean 8, median 7, no mode. 14 to 24: mean 10, median 7, no mode. 14 to 4: mean 6, median 6, mode 4. 14 to 7: mean 6.6, median 7, mode 7 (all by computation). The mean changes every time because it uses every value. The median stayed 7 for 24 and 7 but fell to 6 for 4, because the order changed and a different number became the middle one. The median only cares about the middle; the mode appeared only when a value repeated. Two numbers: change one up by k and another down by k, for example 4 to 6 and 14 to 12; the total stays 40 and the mean stays 8. Look for: totals used, the middle found after re-ordering, and a reason for each change. |

These are the questions on the Think harder, Go further and Finish pages in the workbook. More are in the separate Question Bank.

### 75–82 min Talk time

**DO** Bring the group back to a circle (or camera on, mics on). Ask for volunteers first and choose two or three of them, not only the fastest. Never cold-call a child by name. "Pass" is always fine, and the next person helps. No public tallies, points or rankings.

**SAY** *"Can the mean be a number that nobody in the group has?"*

*"Can the mean be bigger than every value in the list? Why or why not?"*

| Discussion question | Listen for |
|---|---|
| Can the mean be a number that nobody in the group has? Can the mean be bigger than every value in the list? | Yes: the mean of 2 and 3 is 2.5, a number nobody in the group has. No: the mean is always between the smallest and the largest value. Levelling out moves cubes from taller towers to shorter ones, so the level can never be higher than the tallest tower or lower than the shortest. |

### 82–90 min Exit slip, looking back and closing

**DO** Minutes 82 to 86: exit slip. Children write on the exit slip box of the Finish page (or a slip of paper). No talking, no helping. Show the exit ticket slide.

| Exit slip | Answer |
|---|---|
| The mean of 3 numbers is 9. Two of them are 5 and 12. What is the third number? A fourth number, 13, is added. What is the new mean? | 10 (total 3 × 9 = 27; 27 − 5 − 12 = 10). New mean 10 (27 + 13 = 40, 40 ÷ 4 = 10). |

**SAY** *"Today you found means by levelling and by dividing, worked backwards from a mean to a total, and caught a misleading average. And your code added up a whole list for you. Next lesson is our Showcase Challenge: start thinking about a puzzle you would like to turn into a program."*

*"Homework thought for next lesson: decide which starter you will change (A, B, C or Clock Jumper) and what your twist will be, so you can start building straight away."*

## Looking back (2 minutes, minutes 86 to 88)

**SAY** *"Two minutes, just for you, in the "Looking back" box on the Finish page. Finish: something I did well, the hardest part was, and tick what helped you. You can write "pass". If you are still stuck, tick that box and I will check in with you in the open session, in the shared room, with your parent nearby. Nobody marks this and nobody has to show it."*

**DO** Everyone reaches this box: it is on the Finish page, not on the Think harder pages. Collect nothing in front of the group and never read an answer out. You may look at a child's box on camera in the open session if the child offers it. Praise the strategy, not the speed: "you you re-ordered the numbers, and that is what got you unstuck", never "you were quick".

If a child ticks "I am still stuck", check in during the open session the same day: visit the shared breakout room or use a hand signal, with a parent nearby. Never by private message. If a child writes "everything" or "I am bad at maths", say quietly in the open session: "Thank you for telling me. Let's find one small thing that worked today," and point to a specific item they got right. If a child writes "nothing was hard", ask what would make it harder.

**DO** Minutes 88 to 90: stamp the inquiry journal when a child shows you the exit slip. Mark S, D or H in the progress tracker at the back of this guide. If a child marked D or H, plan two minutes of catch-up at the start of the next lesson.

## Extra idea for anyone who wants it

Make a list of 5 whole numbers with mean 10, median 8 and mode 7 (one answer: 7, 7, 8, 13, 15). Then change your program so it also finds the largest value in the list.

## Answer key (workbook)

**Q1** (a) 30 (b) 6 (c) −2, +1, −4, +2, +3; they add up to 0 (the values above and below the mean balance)

**Q2** (a) 68 points (17 × 4) (b) 19 (68 − 15 − 20 − 14) (c) 22 (5 × 18 = 90, 90 − 68 = 22)

**Q3** (a) stays the same: 10 (60 ÷ 6) (b) goes up: 13 (c) goes down: 9 (36 ÷ 4) (d) goes up: 16 (80 ÷ 5)

**Q4** (a) $22 (154 ÷ 7) (b) $15 (c) $12 (d) 6 pupils (e) Not fair: the one $69 pulls the mean up, and 6 of the 7 pupils get less than $22. The median $15 describes a typical pupil better.

**Q5** Trace: i = 1, 2, 3, 4; item i = 5, 9, 6, 12; total = 5, 14, 20, 32. (a) 8 (32 ÷ 4) (b) repeat (length of scores) and say total ÷ (length of scores); it should now say 9 (45 ÷ 5)

**Q6** (a) 31 °C (217 ÷ 7) (b) 3 days (33, 32 and 34) (c) Blank 1: mean. Blank 2: count. It says 3.

**Investigation** Original: mean 8, median 7, no mode. 14 to 24: mean 10, median 7, no mode. 14 to 4: mean 6, median 6, mode 4. 14 to 7: mean 6.6, median 7, mode 7 (all by computation). The mean changes every time because it uses every value. The median stayed 7 for 24 and 7 but fell to 6 for 4, because the order changed and a different number became the middle one. The median only cares about the middle; the mode appeared only when a value repeated. Two numbers: change one up by k and another down by k, for example 4 to 6 and 14 to 12; the total stays 40 and the mean stays 8. Look for: totals used, the middle found after re-ordering, and a reason for each change.

# Lesson 10: Showcase Challenge

Week 5 (second lesson) • Mixed: Maths and Code • Workbook: Lesson 10

| | |
|---|---|
| **Objective** | Use ideas from all nine lessons (sequences, primes, negative numbers, algebra, counting, probability, angles, clock arithmetic, averages) in a mixed challenge round, then plan, build, test and present a small puzzle program, working out the maths by hand first and explaining why it works. |
| **Thinking skills** | Choosing a strategy, connecting topics, planning code, testing with a known answer, presenting and explaining |
| **Materials** | Laptop or desktop with Scratch 3 (scratch.mit.edu), workbook (challenge round on page 1, planning sheet on page 2), pencil, mini whiteboard, the starter projects as .sb3 files (A Mystery Machine, B Prime or Not?, C Lucky Seven, D Clock Jumper; children use File, Load from your computer), or build them from the slides, certificates, stamps. |
| **Online set-up** | Challenge round in teams of 2 in breakout rooms. Building: everyone in the main room with cameras on, asking for help in the chat. Showcase: in two rounds of 4, each child shares their screen for 2 minutes 30 seconds; the audience writes one question for each presenter and the tutor posts stars first (two stars and a question). A child may show a screenshot and the tutor runs the file. Children may paste a screenshot into the chat instead and the tutor runs their file, which removes screen-share switching. Block colours on our slides are simplified; in Scratch, find blocks by the category name on the left, not by colour. |
| **Prepare** | Solve the worksheet yourself first. Set out one materials box per pair. Open the slides for this lesson and check the answers in the speaker notes. |

**DO** Lesson 10 timing: this lesson changes the usual routine, so use this plan instead of the minute marks below. Opening 0 to 2; sneak-peek puzzles 2 to 5; warm-up 5 to 8; challenge round 8 to 22; starter projects 22 to 29; brain break 29 to 32; plan and build 32 to 62 (a working copy of one starter with one change is full marks); showcase 62 to 87 (3 minutes each plus a 20-second switch); exit ticket and certificates 87 to 90. The slide deck has the same times.

### 0–2 min Opening

**SAY** *"Welcome to our last lesson, inquirers! Two parts today. First a challenge round with puzzles from all nine lessons. Then you become the puzzle maker: you build a small program that sets a maths puzzle, and you show it to the group."*

*"Our rule all program long: notice it, test it, explain why. Today you do all three in front of an audience."*

**DO** Greet each child by name. Show the lesson title slide. Read "Today we will…" aloud.

### 2–5 min Sneak-peek puzzles

**SAY** *"Two sneak-peek puzzles. They are ideas we have not learnt yet, so guessing is welcome and "I do not know" is welcome. Nobody is marked. Write your answers on your own paper. When I say "show", everyone holds their paper to the camera at the same moment."*

| Sneak-peek puzzle | Answer and what to listen for |
|---|---|
| Count on in 5s from 4: 4, 9, 14, 19, … Write the next three numbers. | 24, 29 and 34. Any child can count on. |
| The list 4, 9, 14, 19, … carries on forever. Is 2026 in the list? What do you think? How could you tell? | No. Every number in the list ends in 4 or 9 (it is 1 less than a multiple of 5), and 2026 ends in 6. A guess is welcome; today's challenge round uses ideas like this. |

**DO** Puzzle 1 is an access puzzle: any child can try it with objects, a drawing or counting, so everyone gets a first win. Puzzle 2 is a thinking puzzle on the idea of today's lesson.

Papers are shown to the camera by everyone at the same moment, or to you only on camera. Never use private chat and never read wrong answers aloud. Reveal the answers to everyone at minute 5 and say: "Both of these are what we learn today."

Do not record the sneak-peek as a mark. Never announce who got which right, never say who will start on which page, and never use the word "advanced" or "top group".

Invite the WHOLE group, never one child: "When you have finished Start here, you can choose Think harder, Go further or the Investigation." Nobody is singled out. The point printed on page 1 ("After question N you may choose…") is the same for all. Any child can choose it, and any child can go back. Do not split a pair unless both children agree.

### 5–10 min Warm-up

**DO** Lightning relay. The tutor asks a quick question from one of Lessons 1 to 9 and asks for a volunteer; the child who answers asks the next question (from the cards on screen) and invites someone who has not had a turn. Pass is always fine. (5 to 8 minutes. Keep it friendly: no racing or timers, and no points for being first.)

**SAY** *"Lightning relay! I ask a quick question from one of our lessons and ask for a volunteer to answer. The child who answers reads the next card and invites someone who has not had a turn. "Pass" is always fine, and the next person helps. Ready? First one: what is the 10th term of 3, 7, 11, 15?"*

*"Which lesson did that question come from? Can you remember the trick we learned?"*

**DO** At about minute 8, ask the quick questions on the warm-up slides. Use thumbs or a whisper to partners before taking answers. Stop at minute 10.

| Ask | Expect |
|---|---|
| What is the nth term of 4, 7, 10, 13, …? | 3n + 1 |
| Is 91 a prime number? | No: 7 × 13 |
| −5 + 8 − 6 = ? | −3 |
| A regular hexagon: what angle does the turtle turn at each corner? | 60° |

| | |
|---|---|
| Mean of 7, 9, 11 and 13? | 10 |

**10–25 min Explore (in pairs, with objects)**

**SAY** *"Look at the puzzles on page 1. Which one do you think will be hardest? Which lesson does each one come from? Write your guesses at the top of the page."*

**DO** 1. Teams of 2. Open the workbook at page 1: the challenge round.

2. Three puzzles from across the program. Maths first.

3. Score: 1 point for each answer, 1 for a check, 1 for a clear reason. Keep your own score on page 1.

4. After 10 minutes we check together.

**SAY** *"Talk to your partner while you work. I will come round and ask you questions."*

**IF STUCK** Stuck on a puzzle: which lesson does it remind you of? Make a table, list it in order, or draw it.

Got an answer: earn the check point. Check it a different way, or test it in code.

Finished early: write a rule for Q1 that would make 2027 sticks. Is it possible with hexagons?

**DO** Go further (from minute 12, on the explore slide, for any pair that wants it): What if the hexagon puzzle used PENTAGONS joined along one side? Find the rule for the sticks and say how many pentagons use 101 sticks. Answer: 4n + 1 sticks (a pentagon needs 5 sticks and each new one adds 4). 101 sticks: 4n + 1 = 101, so n = 25 pentagons. Nobody has to wait: a pair that has finished can pick this up.

**DO** At about minute 22, say "Pencils down" and bring the group back together. Take the three questions below, one at a time.

| Ask | Expect |
|---|---|
| How did you start the hexagon puzzle? | Make a table of hexagons and sticks; notice + 5 each time; the rule is 5n + 1. |
| How did you know you had found every way to write 40 as two primes? | Go through the primes in order up to 20 and test whether 40 minus each one is prime. |
| Which check did you do in code? | For example, run the hexagon loop, or list the digit-card numbers with nested loops and count the multiples of 3. |

**25–35 min Mini-lesson (you teach, they join in)**

**DO** Show the mini-lesson slides. Think aloud as you work. Say what you notice, what you test and how you check. Children copy how you think.

**SAY** *"Hexagon chains: 6, 11, 16 sticks. Each new hexagon shares one side with the last one, so it adds 5 sticks. The first hexagon is 6 = 5 + 1. So n hexagons use 5n + 1 sticks. For 10 hexagons: 51. I can test that in code: start at 6 and change by 5 nine times."*

*"Mystery Machine: the program picks n, works out 3n + 4 and shows it. If it shows 19, I go backwards: 19 – 4 = 15, 15 ÷ 3 = 5. Before I trust the program, I test it with a number I already know: if n is 5 it must show 19."*

*"Lucky Seven: there are 6 × 6 = 36 equally likely results for two dice. 1 + 6, 2 + 5, 3 + 4, 4 + 3, 5 + 2, 6 + 1 make 7: that is 6 of 36, one sixth. So in 1000 rolls I expect about 167. When I run it I got 152, then 182, then 162: close to 167 but never the same, because the rolls are random."*

| Worked example on the slides | Answer |
|---|---|
| Starter A: Mystery Machine. The program picks n from 1 to 20 and shows 3n + 4. The player works backwards: subtract 4, then divide by 3. Test with a value you know: n = 5 must show 19. | Out 25 → 7. Out 64 → 20. Out 3 is impossible: the smallest OUT for n from 1 to 20 is 7. |
| Starter C: Lucky Seven. Roll two dice 1000 times. Count the 7s. 36 equally likely results; 6 of them make 7. So expect about 1000 ÷ 6. | About 167. Each run gives a slightly different number (for example 152, 182, 162): random results vary. |

**SAY** *"The key idea today is: maths first, then code to test it. Work out the answer by hand first. Then the program is a test: if it agrees with you, you are probably both right; if not, find out which one is wrong. Say it with me."*

**DO** Show the key idea slide at minutes 33 to 35. The "common mistakes" on that slide are for you, not the children. Watch for them during the worksheet:

- Coding before working out the maths, so there is no way to tell whether the output is right.
- Testing only once, or only with the value the program happened to show, instead of a value worked out by hand (and an edge case such as 1 or 0).
- Showcase: reading the code block by block instead of explaining the maths puzzle and why the program works.

**35–40 min Brain break**

**DO** Turtle moves. The tutor calls a polygon; everyone turns on the spot by its outside angle at every corner (triangle: 120°, square: 90°, hexagon: 60°) and counts until they face the front again.

Water. Toilet break for anyone who needs it. Keep it to five minutes.

**40–75 min Worksheet (pairs, then alone, then your choice)**

**DO** Minute 40: give out the workbook. Read the title of Lesson 10 aloud. Say "Pairs first. Talk before you write."

Follow the minute labels on the slides: short teaching slides come before page 1 and before page 2, pairs work together on page 1 (about 40 to 55), each child works alone on page 2 (about 55 to 72), and minutes 72 to 75 are the check together. Any child who has finished can choose Think harder, Go further or the Investigation (all printed in the workbook). The Finish page is for everyone.

**SAY** *"Pairs first. Read each question aloud and talk about it before you write. I am not looking for speed. I am looking for good thinking."*

**DO** Circulate and ask:

- Ask: "What answer should your program give for this input? How do you know?"
- Ask: "What happens with an edge case: 1, 0, or the biggest number?"
- Ask: "In one sentence, what maths is your program about?"

**IF STUCK** Hint 1: "Start from a starter project and change one thing: the rule, the numbers, or the question."

Hint 2: "Write the maths on your planning sheet first. Then find one block for each line of your plan."

Hint 3: "If the program disagrees with you, trace it in a table: which value goes in, what does each block do?"

**DO** Think harder, Go further and the Investigation are open to every child, with no label. The Investigation is printed on the Finish page that every child gets. After question 3 (see the line printed on page 1) any child may choose to carry on or try Think harder. Never announce who is on which page. Do not split a pair unless both children agree.

## Stretch and challenge questions

| Question | Answer |
|---|---|
| Workbook Think harder 1: A row of squares uses 4, 7, 10, 13, ... matchsticks. Which pattern number uses exactly 100 matchsticks? | Pattern 33 (3n + 1 = 100) |
| Workbook Think harder 2: Find all the prime numbers between 80 and 100. | 83, 89 and 97 |
| Workbook Think harder 3: A regular polygon has an angle of 150° inside each corner. How many sides does it have? | 12 (the turn at each corner is 180° − 150° = 30°, and 360 ÷ 30 = 12) |
| Workbook Think harder 4: Cards 1, 2, 3, 4 make 2-digit numbers (no card twice). (a) How many numbers? How many are even? (b) One of the numbers is picked at random. What is the probability it is a multiple of 3? | (a) 12 numbers (4 × 3); 6 are even (3 + 3). (b) 4/12 = 1/3 (12, 21, 24, 42). |
| Workbook Go further 5: Make up a challenge question that mixes TWO topics from the programme (for example primes and the clock, or sequences and negative numbers). Write the question, the answer and the working. Swap with a partner and solve each other's. | Open. Example: "The nth term of a sequence is 7n − 3. Which is the first term that is a multiple of 5?" The terms are 4, 11, 18, 25, so it is the 4th term, 25. Check that the answer is correct and that both topics are really needed. |

| | |
|---|---|
| Workbook Go further 6: A spinner has the numbers 1 to 8, all equally likely. Ravi spins it twice and adds the numbers. What is the probability that the total is 9? Plan a program that tests it 1000 times. | 8/64 = 1/8 (1 + 8, 2 + 7, … , 8 + 1), so about 125 in 1000. Program: set nines to 0; repeat 1000 (if pick random 1 to 8 + pick random 1 to 8 = 9 then change nines by 1); say nines. |
| Workbook Go further 7: The nth term of a sequence is 6n − 1. Is 2027 (the year you start Secondary 1) in the sequence? Which is the first term bigger than 2027? | Yes: it is the 338th term (6 × 338 − 1 = 2027). The first term bigger than 2027 is the 339th, 2033. |
| Workbook Investigation, Finish page (Find all the ways): The sequence 4, 11, 18, 25, … has the rule 7n − 3. Find ALL the prime numbers in this sequence up to 100. Show your list of terms and how you tested each one. Then explain why no term of the sequence is a multiple of 7. First step: write the first 5 terms of the sequence 7n − 3 (4, 11, 18, …). | The terms up to 100 are 4, 11, 18, 25, 32, 39, 46, 53, 60, 67, 74, 81, 88, 95 (14 terms). The primes are 11, 53 and 67 (checked by brute force); the others are even or have a factor 3, 5 or 13 (25 = 5 × 5, 39 = 3 × 13, 81 = 9 × 9, 95 = 5 × 19). No term is a multiple of 7 because 7n is a multiple of 7, and 7n − 3 is 3 less, so dividing by 7 always leaves remainder 4. Look for: the full list, a test for every term using small primes, and the remainder reasoning. Stretch: change the rule to 7n + 1 and repeat. |

These are the questions on the Think harder, Go further and Finish pages in the workbook. More are in the separate Question Bank.

### 75–82 min Talk time

**DO** Bring the group back to a circle (or camera on, mics on). Ask for volunteers first and choose two or three of them, not only the fastest. Never cold-call a child by name. “Pass” is always fine, and the next person helps. No public tallies, points or rankings.

**SAY** *“What was the hardest bug you fixed today? How did you find it?”*

*“Which maths idea from the program would you like to explore further, and what would you try next?”*

| Discussion question | Listen for |
|---|---|
| Showcase: 2 minutes 30 seconds each, in two rounds of 4. Show the puzzle, run the program, show one test where you knew the answer first, and explain the maths that makes it work. | Open answers. Listen for: testing with a value worked out by hand, tracing the code in a table, and maths links (nth term, primes, negative numbers, equations, probability, angles, averages). This is enrichment toward secondary maths and coding, not a preview of exam topics. |

### 82–90 min Exit slip, looking back and closing

**DO** Minutes 82 to 86: exit slip. Children write on the exit slip box of the Finish page (or a slip of paper). No talking, no helping. Show the exit ticket slide.

| Exit slip | Answer |
|---|---|
| | |

| | |
|---|---|
| Write one thing your program does, one test you did (the input and the answer you expected), and the maths that explains it. | Open. Look for a test worked out by hand before running the program, and a correct maths reason. |

**SAY** *"Ten lessons ago you predicted the 100th term of a pattern and checked it in code. Today you built your own puzzle program and explained the maths to an audience. That is what inquirers do. Here is your certificate. Keep noticing, keep testing, and keep asking why!"*

*"Give each child a certificate and tell them one specific thing they did well in their showcase."*

## Looking back (2 minutes, minutes 86 to 88)

**SAY** *"Two minutes, just for you, in the "Looking back" box on the Finish page. Finish: something I did well, the hardest part was, and tick what helped you. You can write "pass". If you are still stuck, tick that box and I will check in with you in the open session, in the shared room, with your parent nearby. Nobody marks this and nobody has to show it."*

**DO** Everyone reaches this box: it is on the Finish page, not on the Think harder pages. Collect nothing in front of the group and never read an answer out. You may look at a child's box on camera in the open session if the child offers it. Praise the strategy, not the speed: "you you broke the problem into smaller parts, and that is what got you unstuck", never "you were quick".

If a child ticks "I am still stuck", check in during the open session the same day: visit the shared breakout room or use a hand signal, with a parent nearby. Never by private message. If a child writes "everything" or "I am bad at maths", say quietly in the open session: "Thank you for telling me. Let's find one small thing that worked today," and point to a specific item they got right. If a child writes "nothing was hard", ask what would make it harder.

**DO** Minutes 88 to 90: stamp the inquiry journal when a child shows you the exit slip. Mark S, D or H in the progress tracker at the back of this guide. If a child marked D or H, plan two minutes of catch-up at the start of the next lesson.

## Extra idea for anyone who wants it

Level up your program: ask 5 questions and keep a score, let the player choose the rule, or count every dice total in a list. Write a Python version and check it gives the same results.

## Answer key (workbook)

**Q1** (a) 51 (b) $5n + 1$ (c) 20 hexagons ($5n + 1 = 101$) (d) 76: it checks a chain of 15 hexagons ($6 + 14 \times 5$)

**Q2** (a) 49 (factors 1, 7, 49; the only other 2-digit number with exactly 3 factors is 25) (b) $2 \times 2 \times 3 \times 7$ (c) 3 + 37, 11 + 29, 17 + 23 (3 ways)

**Q3** (a) −7 °C ($-4 + 9 = 5$, $5 - 12 = -7$) (b) 5 ($5 \times 3 - 10 = 5$) (c) 15 sides ($360 \div 24$) (d) 1 o'clock ($9 + 100 = 109 = 9 \times 12 + 1$) (parts e and f are now in Think harder 4)

**Q4** (a) 31; 18 ($(58 - 4) \div 3$) (b) 4 factors (1, 7, 13, 91): not prime (c) about 139: 5 of the 36 equally likely results make 8 (2 + 6, 3 + 5, 4 + 4, 5 + 3, 6 + 2), and $1000 \times 5 \div 36 \approx 139$; the program gives a number near this, not exactly

**Q5** Open: an idea, the puzzle, its answer worked out by hand, and a twist.

**Q6** Open: blocks (under 15) and at least two tests worked out before running, one of them an edge case.

**Investigation** The terms up to 100 are 4, 11, 18, 25, 32, 39, 46, 53, 60, 67, 74, 81, 88, 95 (14 terms). The primes are 11, 53 and 67 (checked by brute force); the others are even or have a factor 3, 5 or 13 (25 = 5 × 5, 39 = 3 × 13, 81 = 9 × 9, 95 = 5 × 19). No term is a multiple of 7 because 7n is a multiple of 7, and 7n − 3 is 3 less, so dividing by 7 always leaves remainder 4. Look for: the full list, a test for every term using small primes, and the remainder reasoning. Stretch: change the rule to 7n + 1 and repeat.

## Progress tracker

After each exit ticket, write S (secure: correct and can explain), D (developing: correct but cannot explain yet) or H (needs help). Use the notes for what to revisit next lesson.

| Student | L1 | L2 | L3 | L4 | L5 | L6 | L7 | L8 | L9 | L10 | Notes |
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### Ending the program

- In Lesson 10, give each child a certificate and their completed inquiry journal.
- Send parents a short message through the approved channel: one thing their child did well, and one thing to keep practising.
- Delete or hand over all Student Data within 7 days after the last lesson (this is in the tutor agreement).

# Tutor checklist

## Before each lesson

- Read the lesson plan and solve the worksheet yourself without looking at the key.
- Check materials are ready, or the online set-up has been sent.
- Test the platform, screen share and annotation tools 10 minutes early (online).
- Print or open the slide deck for the lesson.

## During the lesson

- Circulate and ask questions rather than giving answers.
- Use only the approved channels to contact parents. No private messages to children.
- Do not take photos or recordings unless the Company has approved it and parents have consented.
- If you have a concern about a child's safety or welfare, tell the Company straight away.

## After the lesson

- Stamp inquiry journals and fill in the tracker.
- Collect worksheets and answer keys (face to face) and keep them secure.
- Note any child who needs help, and plan a short catch-up at the start of the next lesson.

# Math Inquirers Holiday Program

*Notice it. Test it. Explain why.*

### A 10-lesson holiday maths program for Primary 6 students after the PSLE

Your child will spend the holidays as a young mathematician: noticing patterns, testing ideas with real objects and explaining their thinking. Lessons are live online in small groups, hands-on, and built around thinking, not speed.

### What your child will do

- Lesson 1, Code the Pattern: sequences and the nth term
- Lesson 2, Prime Hunters: primes, factors and prime factorisation
- Lesson 3, Below Zero: negative numbers
- Lesson 4, Algebra Machines: expressions and equations
- Lesson 5, Every Possible Way: counting
- Lesson 6, Fair or Unfair?: probability
- Lesson 7, Turtle Geometry: angles in polygons
- Lesson 8, Secret Codes: clock (modular) arithmetic
- Lesson 9, Data Detectives: averages and data
- Lesson 10, Showcase Challenge: mixed: maths and code

### How you can help

- Ask “How do you know?” when your child tells you an answer.
- Let your child try first. Please sit nearby, but let them find the answer.
- Print the workbook and cut-outs, and keep scissors, a ruler and a pencil ready.

### Practical details

| | |
|---|---|
| **Dates** | [start date] to [end date], two lessons a week |
| **Time** | [day] and [day], [start time] to [end time] |
| **Where** | Live online video class (link sent after registration) |
| **Group size** | [number] children |
| **Fee** | S$500 introductory (regular S$650), workbook included, payable in 3 instalments |
| **Contact** | [name, phone or email] |

*Math Inquirers Holiday Program is an independent enrichment program. It is inspired by thinking-skills enrichment ideas but is not affiliated with, run by or endorsed by the Ministry of Education or any school.*